A Novel Iterative Self-Organizing Pixel Matrix Entanglement Classifier for Remote Sensing Imagery

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***Abstract*— The previously presented self-organizing pixel entanglement neural network (SOPENN) model only establishes 2-D basis vectors that are orthogonal to each other in the Hilbert space, which cannot sufficiently reflect the spectral information and the entanglement characteristics of pixels in multispectral images. Therefore, an iterative self-organizing pixel matrix entanglement (ISOPME) image classification model is proposed in this article. Quantum pixel matrix entanglement (PME) is based on quantum pixel entanglement, which considers a pixel as a quantum, and the quantum entanglement theory in quantum informatics is applied in the PME. First, the PME theory was developed to associate the quantum states of pixels with their gray values. Second, the PME coefficient was proposed to extract the entanglement relationship between pixel matrices with 3-D basis vectors in the Hilbert space. Finally, an ISOPME model was developed to implement a self-organizing clustering for multispectral remote sensing image classification. The experimental results for four test areas demonstrate that: 1) the proposed ISOPME approach achieves an average classification accuracy of 92.02% and a Kappa coefficient (KC) of 0.88; 2) when compared with four traditional unsupervised classification methods, ISOPME on average improves the classification accuracy by 8.96% and the KC by 0.14; and 3) the classification accuracy and KC from ISOPME reach the same level as the more sophisticated supervised classification methods, such as support vector machine (SVM), and are close to those obtained using deep learning (DL) classification methods.**

***Index Terms*— Classification, pixel matrix entanglement (PME), quantum entanglement, remote sensing images, self-organizing clustering.**

I. Introduction

SINCE its discovery, quantum entanglement has promised to revolutionize computation, communication, imaging, and other applications. Zhou et al. [1] pioneered the study on pixel entanglement theory, named "self-organizing pixel entanglement neural network (SOPENN)," for the classification of remotely sensed images (RSIs). However, the SOPENN model developed by Zhou et al. [1] is only established on 2-D basis vectors orthogonal to each other in the Hilbert space; therefore, it is challenging to guarantee high accuracy values for RSI classification. While quantum pixel systems are indeed entangled with each other in a multidimensional array space, this implies that the entangled state of a quantum pixel system is often a mixed state, which contains pixels of multiple types of ground objects. This article presents a new model that considers the relationship between spectral information and entanglement characteristics of pixels.

RSI classification methods can be roughly divided into supervised and unsupervised classification and shallow and deep learning (DL) methods [2], [3], [4]. Shallow supervised classification methods include but are not limited to the minimum distance classifier (MDC) [5], the maximum likelihood classifier (MLC) [6], support vector machines (SVMs) [7], artificial neural networks (ANNs) [8], and decision trees (DTs) [9]. Unsupervised classification methods include but are not limited to the iterative self-organizing data analysis (ISODATA) [10], K-Means [11] clustering approaches, self-organizing mapping (SOM) networks [12], and fuzzy C-means (FCM) clustering [13]. Supervised DL classification methods include but are not limited to convolutional neural networks (CNNs) [14], deep belief networks (DBNs) [15], stacked autoencoders (SAEs) [16], and sparse coding networks (SCNs) [17]. Selecting the most suitable RSI classification method requires a comprehensive consideration of multiple factors such as the number and quality of training samples, the requirements for classification accuracy, and the capacity of the computing platforms on which the classification is performed [18], [19]. Various classification methods and algorithms have been designed to improve the classification accuracy.

Among the unsupervised classification methods mentioned above, ISODATA is one of the most widely used since this method can dynamically adjust the clustering center by automatically performing "merging" and "splitting" of clusters and continuously absorb intermediate experience to update clustering results until all objects in the image are completely classified [20]. Rydberg and Borgefors [21] combined the ISODATA algorithm with an edge detection algorithm for

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farmland boundary delineation, which improved the efficiency of the original operations [21]. Zhong et al. [22] simulated a vertebrate immune system and integrated it with unsupervised clustering classifiers such as ISODATA and K-Means to propose a more accurate unsupervised artificial immune classifier. Tarabalka et al. [23] proposed an image clustering technique that combines the ISODATA algorithm and the Gaussian method, comprehensively utilizing spectral and spatial features and acquiring excellent classification accuracy. Long and Singh [24] proposed an entropy-based multispectral image classification algorithm that achieved a higher accuracy and robustness than the ISODATA algorithm. Jiao et al. [25] proposed an unsupervised spectral matching classifier based on artificial DNA computation, which obtained higher classification accuracy than K-Means, ISODATA, FCM, and other clustering classifiers. Ulug and Karslioglu [26] believed that ISODATA is a heuristic algorithm capable of automatic clustering based on user-defined rules. Zhang et al. [27] utilized four supervised methods, including SVM and random forest (RF), and two unsupervised methods, the hotspot analysis and ISODATA algorithm, to detect the canopy of banana wilt infections. Although the ISODATA algorithm is widely used as an effective image classification method, there is room for improvement. For example, the initial cluster centers of the ISODATA algorithm must be selected manually, the number of iterations must be set manually, and the clustering termination must also be intervened manually. In addition, the traditional ISODATA algorithm, as well as K-Means, SOM, and FCM, only considers the Euclidean distance between the pixels of the classified image without fully considering the rich spectral information of the RSIs, making it challenging to guarantee the classification accuracy. Based on the reasons above, this article employs a quantum entanglement theory to extract the entanglement characteristics of pixel matrices hidden in the pixel level of RSI, namely, pixel matrix entanglement (PME).

A few scholars in the remote sensing community have attempted to improve classification methods by incorporating the quantum mechanics theory based on the advantages of quantum entanglement characteristics, such as action at a distance, strong correlation of quantum particles, and universal entanglement of quantum states [28], [29], [30], [31], [32]. For example, Ruan et al. [33] proposed a global quantum feature extraction method based on the Schmidt decomposition and a quantum learning algorithm for classification based on the Hamming distance of image features, leading to the effective classification of large-scale images. Chen et al. [34] developed a quantum genetic-optimized sparse representation method to propose a hyperspectral remote sensing image classification method, QGASR-CNN, which effectively addresses small samples and reduces salt-pepper errors, thus achieving improved accuracy in hyperspectral RSI classification. Zhou et al. [1] proposed an SOPENN algorithm for RSI classification that acquires higher accuracy than traditional unsupervised classification methods. Vernuccio et al. [35] merged quantum optics with artificial intelligence, exploring their applications in image classification, spectrum recognition, and other fields. Sebastianelli et al. [36] proposed a novel hybrid quantum CNN (QCNN) image classifier that was successfully applied to land use and land cover image classification. Atchade-Adelomou and Alonso-Linaje [37] proposed a hybrid image classification model that combines quantum and classical computing to comprehensively utilize convolutional networks and variational quantum filters]. Easom-McCaldin et al. [38] proposed a deep quantum neural network based on single-qubit encoding for image classification, which inherits the advantages of traditional CNN technology while reducing the number of parameters. Zeguendry et al. [39] compared the classification accuracy of a variational quantum classifier with that of classical classifiers based on quantum machine learning theory.

The classification methods of RSI based on the principles of quantum mechanics can sufficiently exploit the quantum properties of objects. Therefore, this article considers mixed pixels as quantum pixel matrices with quantum states, most of which are entangled, allowing the entanglement characteristics of the quantum pixel matrices to be leveraged to achieve accurate image classification. As reviewed earlier, the previous method taking advantage of the pixel entanglement characteristics was established only for mutually orthogonal 2-D basis vectors in the Hilbert space, which did not fully explore the relationship between the spectral information and the entanglement characteristics of pixel particles. Thus, this article presents a novel iterative self-organizing PME (ISOPME) model that overcomes these shortages, improving RSI classification.

II. Theory of Quantum PME

A. Quantum Pixel Matrix Expression

Any RSI is composed of quantized pixels (named "quantum pixels") belonging to different objects, which are entangled with each other in a multidimensional array form. To describe the proposed method, we consider a color image with three primary colors (red, green, and blue, see Fig. 1) as an example. The pixels of this image can be quantized as

$$|\phi_p\rangle = \alpha|1\rangle + \beta|2\rangle + \gamma|3\rangle \tag{1}$$

where $|\phi_p\rangle$ is the quantum state of the quantum pixel p; $|1\rangle$, $|2\rangle$, and $|3\rangle$ are the orthogonal eigenvectors of $|\phi_p\rangle$; and α, β, and γ are the eigenvalues of $|\phi_p\rangle$, which satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1. \tag{2}$$

The relationship between the eigenvalues of the pixel quantum states and the pixel gray values is expressed as

$$\alpha = \cos\left(\frac{G_R}{255}\right), \quad \beta = \cos\left(\frac{G_G}{255}\right), \quad \gamma = \cos\left(\frac{G_R}{255}\right) \tag{3}$$

where G_R, G_G, and G_B are the gray values of the red (R), green (G), and blue (B) bands, respectively. Consequently, the color image can be represented as a pixel matrix with a superposition of multiple quantum pixels, as shown in Fig. 1.

If the size of a single-band image is $m \times n$ and all pixels in the single-band image are quantized, the corresponding

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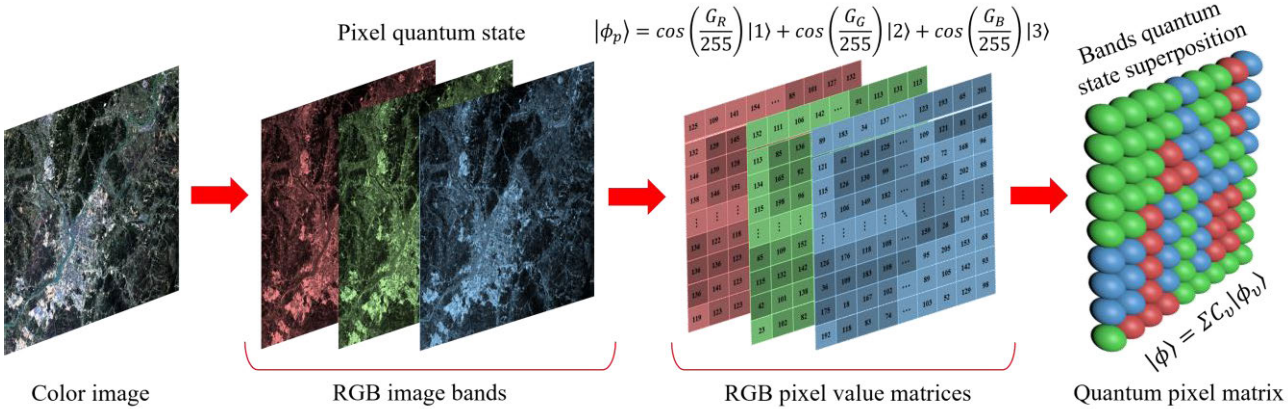


Fig. 1. Pixel matrix expression for a color RSI.

quantum pixel matrix is expressed as

$$|\phi_1\rangle = \begin{bmatrix} |\phi\rangle_{1,1} & |\phi\rangle_{1,2} & \cdots & |\phi\rangle_{1,n} \\ |\phi\rangle_{2,1} & |\phi\rangle_{2,2} & \cdots & |\phi\rangle_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ |\phi\rangle_{m,1} & |\phi\rangle_{m,2} & \cdots & |\phi\rangle_{m,n} \end{bmatrix}. \tag{4}$$

For the pixel in the R-band from the color image, the quantum state is $|\phi_{p1}\rangle = \alpha_1|1\rangle + \beta_1|2\rangle + \gamma_1|3\rangle$, the quantum state in the G-band is $|\phi_{p2}\rangle = \alpha_2|1\rangle + \beta_2|2\rangle + \gamma_2|3\rangle$, and the quantum state in the B-band is $|\phi_{p3}\rangle = \alpha_3|1\rangle + \beta_3|2\rangle + \gamma_3|3\rangle$. The quantum state $|\phi\rangle$ of each pixel in the color image is considered as the superposition state of corresponding quantum pixels in three bands [40], [41], [42] so that $|\phi\rangle$ contains the spectral information of each band of color image, that is,

$$|\phi\rangle = C_1|\phi_{p1}\rangle + C_2|\phi_{p2}\rangle + C_3|\phi_{p3}\rangle \tag{5}$$

where C_1–C_3 are the eigenvalues of $|\phi_{p1}\rangle$–$|\phi_{p3}\rangle$, respectively. Therefore, (5) can also be expressed as

$$|\phi\rangle = (\alpha_1 + \alpha_2 + \alpha_3)|1\rangle + (\beta_1 + \beta_2 + \beta_3)|2\rangle + (\gamma_1 + \gamma_2 + \gamma_3)|3\rangle. \tag{6}$$

B. Theory on PME

Assume that an $m \times n$ sized quantum pixel matrix A, expressed by $|\phi_A\rangle = |\phi_A\rangle_{i,j} \in C^{m\times n}$ and a $p \times q$ sized quantum pixel matrix B, expressed by $|\phi_B\rangle = |\phi_B\rangle_{i,j} \in C^{p\times q}$, i.e.,

$$|\phi_A\rangle = \begin{bmatrix} |\phi_A\rangle_{1,1} & |\phi_A\rangle_{1,2} & \cdots & |\phi_A\rangle_{1,n} \\ |\phi_A\rangle_{2,1} & |\phi_A\rangle_{2,2} & \cdots & |\phi_A\rangle_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ |\phi_A\rangle_{m,1} & |\phi_A\rangle_{m,2} & \cdots & |\phi_A\rangle_{m,n} \end{bmatrix} \tag{7}$$

$$|\phi_B\rangle = \begin{bmatrix} |\phi_B\rangle_{1,1} & |\phi_B\rangle_{1,2} & \cdots & |\phi_B\rangle_{1,q} \\ |\phi_B\rangle_{2,1} & |\phi_B\rangle_{2,2} & \cdots & |\phi_B\rangle_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ |\phi_B\rangle_{p,1} & |\phi_B\rangle_{p,2} & \cdots & |\phi_B\rangle_{p,q} \end{bmatrix}. \tag{8}$$

Then, $|\phi_A\rangle\langle\phi_B|$ is the tensor product of pixel matrix A and pixel matrix B, also known as the Kronecker product, and results into an $mp \times nq$ sized block matrix, denoted as follows:

$$|\phi_A\rangle\langle\phi_B| = \begin{bmatrix} |\phi_A\rangle_{1,1} & |\phi_A\rangle_{1,2} & \cdots & |\phi_A\rangle_{1,n} \\ |\phi_A\rangle_{2,1} & |\phi_A\rangle_{2,2} & \cdots & |\phi_A\rangle_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ |\phi_A\rangle_{m,1} & |\phi_A\rangle_{m,2} & \cdots & |\phi_A\rangle_{m,n} \end{bmatrix} \otimes \begin{bmatrix} \langle\phi_B|_{1,1} & \langle\phi_B|_{1,2} & \cdots & \langle\phi_B|_{1,q} \\ \langle\phi_B|_{2,1} & \langle\phi_B|_{2,2} & \cdots & \langle\phi_B|_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ \langle\phi_B|_{p,1} & \langle\phi_B|_{p,2} & \cdots & \langle\phi_B|_{p,q} \end{bmatrix} \tag{9}$$

where the quantum state $|\phi_A\rangle_{i,j}$ $(i = 1, 2, \ldots, m; j = 1, 2, \ldots, n)$ and the quantum state $|\phi_B\rangle_{i,j}$ $(i = 1, 2, \ldots, p; j = 1, 2, \ldots, q)$ are the quantum pixels of A and B, respectively. As shown in Fig. 2, the pixel matrix A and the pixel matrix B are generally mixed regions with multiple ground objects.

The quantum pixel matrix of $|\phi_A\rangle\langle\phi_B|$ is as follows:

$$|\phi_A\rangle\langle\phi_B| = \begin{bmatrix} |\phi_A\rangle_{1,1}\langle\phi_B| & |\phi_A\rangle_{1,2}\langle\phi_B| & \cdots & |\phi_A\rangle_{m,1}\langle\phi_B| \\ |\phi_A\rangle_{2,1}\langle\phi_B| & |\phi_A\rangle_{2,2}\langle\phi_B| & \cdots & |\phi_A\rangle_{m,1}\langle\phi_B| \\ \vdots & \vdots & \ddots & \vdots \\ |\phi_A\rangle_{m,1}\langle\phi_B| & |\phi_A\rangle_{m,2}\langle\phi_B| & \cdots & |\phi_A\rangle_{m,n}\langle\phi_B| \end{bmatrix}. \tag{10}$$

Suppose that the quantum pixels in a pixel matrix are represented by (1) and (5), and the quantum state of τ pixels in pixel matrices A and B can be represented as (11); then, the entanglement occurs

$$|\phi_\tau\rangle = \lambda_1|1, \ldots, 1\rangle_\tau + \lambda_2|2, \ldots, 2\rangle_\tau + \lambda_3|3, \ldots, 3\rangle_\tau \tag{11}$$

where $|\phi_\tau\rangle$ represents the entangled state of τ pixels; $|1\rangle$–$|3\rangle$ are orthogonal eigenvectors of $|\phi_\tau\rangle$; $|1, \ldots, 1\rangle_\tau$ is the tensor product of $|1\rangle$; $|2, \ldots, 2\rangle_\tau$ is the tensor product of $|2\rangle$; $|3, \ldots, 3\rangle_\tau$ is the tensor product of $|3\rangle$; and λ_1–λ_3 are the eigenvalues of $|\phi_\tau\rangle$ that satisfy the normalization condition

$$|\lambda_1|^2 + |\lambda_2|^2 + |\lambda_3|^2 = 1. \tag{12}$$

For the quantum state of composite pixel matrix AB composed of pixel matrices A and B, the density operator ρ can be

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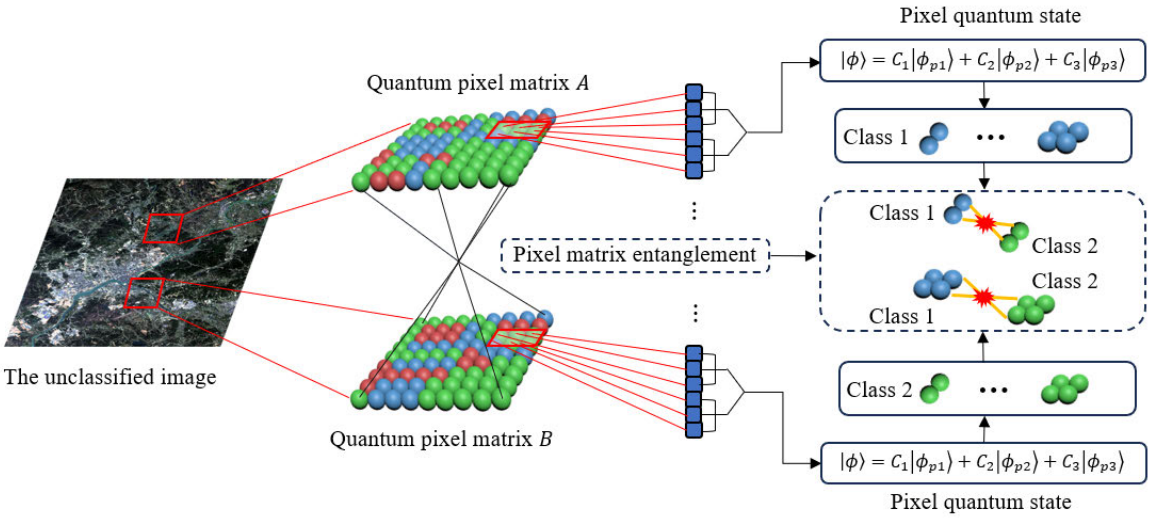


Fig. 2. PME. The pixel matrices A and B are generally mixed regions that is composed of multiple object classes from the same image. The quantum states of pixel matrices are the entangled states.

used to describe the entanglement relationship in the Hilbert space [43], [44], [45] as follows:

$$\rho_\tau = \sum_\tau p_\tau |\phi_\tau\rangle\langle\phi_\tau| \tag{13}$$

where p_τ is the probability of $|\phi_\tau\rangle$, which describes the measurement probability of entanglement occurring of the composite pixel matrix AB in the Hilbert space. Furthermore, if the composite pixel matrix AB is measured, the probability of $|\phi_\tau\rangle$ is

$$p(\tau) = \langle\phi|M_\tau^\dagger M_\tau|\phi\rangle \tag{14}$$

where M_τ is the measurement operator of $|\phi_\tau\rangle$ and $M_\tau^\dagger$ is the Hermitian conjugate of M_τ, satisfying the relationship of $\sum M_\tau^\dagger M_\tau = I$, that is,

$$\sum_\tau \langle\phi|M_\tau^\dagger M_\tau|\phi\rangle = \sum_\tau p_\tau = 1 \tag{15}$$

where $\langle\phi|M_\tau^\dagger M_\tau|\phi\rangle$ is the inner product between $|\phi\rangle$ and $M_\tau^\dagger M_\tau|\phi\rangle$. The entangled state of the measured composite pixel matrix AB is

$$|\phi_{AB}\rangle = p_A|\phi_A\rangle + p_B|\phi_B\rangle = \frac{M_r|\phi\rangle}{\sqrt{\langle\phi|M_r^\dagger M|\phi\rangle}}. \tag{16}$$

In this case, the pixel matrices A and B exhibit entanglement correlations. Conversely, if the quantum state of the composite pixel matrix AB cannot be represented by (16), then its quantum state is not an entangled state, but a disentangled state (also known as "*separable state*") [45], which means that the PME does not occur.

For the disentangled state, the density operator ρ of the composite pixel matrix is

$$\rho = \sum_\tau p_\tau \rho_1 \otimes \rho_2 \otimes \cdots \otimes \rho_\tau; \quad \sum_\tau p_\tau = 1. \tag{17}$$

The measurement probability of the quantum states occurring in the Hilbert space can be understood as the quantum nonlocality (also named "*Bell nonlocality*") [46], which possesses action at a distance, that is, the nonlocal interaction between two quantum states located in two noncontiguous regions of the space. The quantum nonlocality also exists in the entangled state. When the entangled pixel matrix AB is measured, if the entangled state of pixel matrix A spins from up to down and the entangled state of pixel matrix B spins from down to up, then the entangled state of the pixel matrix AB is

$$|\phi'_{AB}\rangle = \frac{\sqrt{2}}{2}\left(\left|\phi_A^\uparrow\right\rangle\left\langle\phi_B^\downarrow\right| - \left|\phi_A^\downarrow\right\rangle\left\langle\phi_B^\uparrow\right|\right). \tag{18}$$

Substituting the entanglement relationship of τ pixels from (11) into (13), we obtain

$$\begin{aligned}\rho_\tau = &\left(\lambda_1|1,\ldots,1\rangle_\tau + \lambda_2|2,\ldots,2\rangle_\tau + \lambda_3|3,\ldots,3\rangle_\tau\right)\\ &\otimes\left(\overline{\lambda_1}\langle 1,\ldots,1|_\tau + \overline{\lambda_2}\langle 2,\ldots,2|_\tau + \overline{\lambda_3}\langle 3,\ldots,3|_\tau\right)\end{aligned} \tag{19}$$

where $\overline{\lambda_1}$, $\overline{\lambda_2}$, and $\overline{\lambda_3}$ and λ_1, λ_2, and λ_3 are complex conjugate to each other, and $|\phi_\tau$ and $|\phi_\tau\rangle$ are mutually dual. Assuming that the quantum state of pixel matrix A is $|\phi_A\rangle = a|1\rangle + b|2\rangle + c|3\rangle$ and the quantum state of pixel matrix B is $|\phi_B\rangle = x|1\rangle + y|2\rangle + z|3\rangle$, the density operators for pixel matrices A and B are

$$\rho_A = |\phi_A\rangle\langle\phi_A| = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \otimes \begin{bmatrix} \bar{a} & \bar{b} & \bar{c} \end{bmatrix} = \begin{bmatrix} a\bar{a} & a\bar{b} & a\bar{c} \\ b\bar{a} & b\bar{b} & b\bar{c} \\ c\bar{a} & c\bar{b} & c\bar{c} \end{bmatrix} \tag{20}$$

$$\rho_B = |\phi_B\rangle\langle\phi_B| = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \otimes \begin{bmatrix} \bar{x} & \bar{y} & \bar{z} \end{bmatrix} = \begin{bmatrix} x\bar{x} & x\bar{y} & x\bar{z} \\ y\bar{x} & y\bar{y} & y\bar{z} \\ z\bar{x} & z\bar{y} & z\bar{z} \end{bmatrix}. \tag{21}$$

If the quantum states of pixel matrix A and pixel matrix B are mixed into a new quantum state, the "*mixed state*" $|\phi_{AB}\rangle$, the relevant quantum pixel matrix AB is viewed as a composite system of quantum pixel matrices A and B, and the density

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operator ρ_{AB} of $|\phi_{AB}\rangle$ can be represented by

$$\rho_{AB} = |\phi_{AB}\rangle\langle\phi_{AB}| = \begin{bmatrix} a\bar{a} & a\bar{b} & a\bar{c} \\ b\bar{a} & b\bar{b} & b\bar{c} \\ x\bar{a} & c\bar{b} & c\bar{c} \end{bmatrix} \otimes \begin{bmatrix} x\bar{x} & x\bar{y} & x\bar{z} \\ y\bar{y} & y\bar{y} & y\bar{z} \\ z\bar{y} & z\bar{y} & z\bar{z} \end{bmatrix}. \tag{22}$$

During the composition of the quantum pixel matrix AB, the density operator of pixel matrix A generates a corresponding reduction [47], [48], [49], [50], i.e.,

$$\rho^A = \mathrm{tr}_B(\rho_{AB}) \tag{23}$$

where ρ^A is the reduced density operator of pixel matrix A in the process of constituting the composite pixel matrix AB, and tr_B is the trace of pixel matrix B, which can also be considered as a mapping function [51], i.e.,

$$\mathrm{tr}_B = (|a_1 b_1\rangle\langle a_2 b_2|) \equiv |a_1\rangle\langle a_2|\,\mathrm{tr}(|b_1\rangle\langle b_2|) \tag{24}$$

where $|a_1\rangle$ and $|a_2\rangle$ are the quantum states of pixel matrix A, $|b_1\rangle$ and $|b_1\rangle$ are the quantum states of pixel matrix B, $|b_1\rangle\langle b_2|$ is a square matrix, and $\mathrm{tr}(|b_1\rangle\langle b_2|)$ is the sum of the diagonal elements of $|b_1\rangle\langle b_2|$.

Moreover, the entangled states can be either pure or mixed [52], [53], [54]. The quantum entropy [55] is introduced to quantitatively analyze the size of the PME between quantum pixel matrices A and B

$$S(\rho) = -\mathrm{tr}\left(\rho \log_2 \rho\right) = -\sum i\lambda_i \log_2 \lambda_i \tag{25}$$

where ρ is the density operator of the composite quantum matrix AB and λ_i is the eigenvalue of ρ.

The entangled state $|\phi_\tau\rangle$ and density operator ρ_τ in (11) and (13) make it impossible to arbitrarily select the quantum state of any pixel matrix for determination; therefore, further clarification is required on the entanglement characteristics of the relevant quantum pixel matrices [53]. This article proposes that the composite quantum pixel matrix AB composed of τ quantum pixels has the following entanglement relationship to explore the entanglement characteristics of pixel matrices in RSIs and further determine whether the entangled state of the quantum pixel matrix is pure or mixed

$$\rho_{i_1,\ldots,i_\tau} = \left(\frac{|1,\ldots,1\rangle_\tau + |2,\ldots,2\rangle_\tau + |3,\ldots,3\rangle_\tau}{\sqrt{3}}\right) \otimes \left(\frac{\langle 1,\ldots,1|_\tau + \langle 2,\ldots,2|_\tau + \langle 3,\ldots,3|_\tau}{\sqrt{3}}\right). \tag{26}$$

According to (23), for the quantum pixel $C_i \in \{C_{i_1},\ldots,C_{i_\tau}\}$, the reduced density operator of the quantum pixel C_i while forming the composite quantum pixel matrix AB is

$$\rho^i = \mathrm{tr}_{i_1,\ldots,i_\tau - i}\left(\rho_{i_1,\ldots,i_\tau}\right) = \frac{|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3|}{3}. \tag{27}$$

In the above equations, the subscript $i_1,\ldots,$ and $i_\tau - i$ represents the fact that the quantum pixel $C_i (i = 1, 2, \ldots, \tau)$ is removed from the entire quantum pixel set $\{C_{i_1},\ldots,C_{i_\tau}\}$.

From an analysis of (26) and (27), we conclude that

$$\begin{aligned} &\mathrm{tr}\left(\rho^2_{i,\ldots,i}\right) \\ &= \mathrm{tr}\left(\left(\left(\frac{|1,\ldots,1\rangle_\tau + |2,\ldots,2\rangle_\tau + |3,\ldots,3\rangle_\tau}{\sqrt{3}}\right) \otimes \left(\frac{\langle 1,\ldots,1|_\tau + \langle 2,\ldots,2|_\tau + \langle 3,\ldots,3|_\tau}{\sqrt{3}}\right)\right)^2\right) \\ &= \frac{1}{9}\,\mathrm{tr}\left((|1,\ldots,1\rangle + |2,\ldots,2\rangle + |3,\ldots,3\rangle)^2 \otimes (\langle 1,\ldots,1| + \langle 2,\ldots,2| + \langle 3,\ldots,3|)^2\right) \\ &= 1 \end{aligned} \tag{28}$$

$$\begin{aligned} &\mathrm{tr}\left(\left(\rho^i\right)^2\right) \\ &= \mathrm{tr}\left(\left(\frac{|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3|}{3}\right)^2\right) \\ &= \frac{1}{9}\,\mathrm{tr}\left((|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3|)^2\right) \\ &= \frac{1}{3}. \end{aligned} \tag{29}$$

With (28) and (29), it can be analyzed that a quantum pixel matrix has the following entanglement characteristics.

1) When $\mathrm{tr}(\rho^2) = 1$, the entangled state of the quantum pixel matrix is a pure state.
2) When $\mathrm{tr}(\rho^2) < 1$, the entangled state of the quantum pixel matrix is a mixed state.

Therefore, it can be concluded that the entangled state of the composite quantum pixel matrix AB is a pure state, whereas the entangled states of the quantum pixel matrices A and B that constitute the composite quantum pixel matrix AB are mixed states.

The entanglement relationship of the composite pixel matrix AB with a pure state can be determined by the quantum entropy of pixel matrices A and B [56]

$$E(|\phi\rangle_{AB}) = S(\rho_A) - S(\rho_B) \tag{30}$$

where $E(|\phi\rangle_{AB})$ is the entanglement measure of the composite pixel matrix AB; and ρ_A and ρ_B, the reduced density operators for quantum pixel matrices A and B, are also named "*reduced density matrices*" [53], [54].

With (27), it can also be analyzed that the entangled state of the quantum pixel $C_i \in \{C_{i_1},\ldots,C_{i_\tau}\}$ is a mixed state of $|1\rangle\langle 1|$, $|2\rangle\langle 2|$, and $|3\rangle\langle 3|$ with a probability of 1/3 each. Thus, (13) indicates that the entangled state of the quantum pixel C_i can be obtained by

$$\rho^i = \left|\lambda_1\right|^2 |1\rangle\langle 1| + \left|\lambda_2\right|^2 |2\rangle\langle 2| + \left|\lambda_3\right|^2 |3\rangle\langle 3|. \tag{31}$$

For composite quantum pixel matrix AB in (27), the density operator of the entangled state has the following relationship:

$$\rho_{i_1,\ldots,i_\tau}\left(\frac{|1,\ldots,1\rangle + |2,\ldots,2\rangle + |3,\ldots,3\rangle}{\sqrt{3}}\right) = \left(\frac{|1,\ldots,1\rangle + |2,\ldots,2\rangle + |3,\ldots,3\rangle}{\sqrt{3}}\right). \tag{32}$$

Therefore, it can be found that the eigenvector of $\rho_{i_1,\ldots,i_\tau}$ is equal to $((|1,\ldots,1\rangle + |2,\ldots,2\rangle + |3,\ldots,3\rangle)/\sqrt{3})$, the

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eigenvalue λ is 1, and the corresponding trace $\mathrm{tr}(\rho)$ is 1. With (25), it can be obtained that the quantum entropy $S(\rho_{AB})$ of the composite quantum pixel matrix AB is zero.

Similarly, for quantum pixel matrices A and B, which constitute the composite quantum pixel matrix AB, the entangled states have the following characteristics:

$$\begin{aligned}\rho^i|1\rangle &= \frac{|1\rangle\langle 1|+|2\rangle\langle 2|+|3\rangle\langle 3|}{3}|1\rangle = \frac{|1\rangle}{3}\\ \rho^i|2\rangle &= \frac{|1\rangle\langle 1|+|2\rangle\langle 2|+|3\rangle\langle 3|}{3}|2\rangle = \frac{|2\rangle}{3}\\ \rho^i|3\rangle &= \frac{|1\rangle\langle 1|+|2\rangle\langle 2|+|3\rangle\langle 3|}{3}|3\rangle = \frac{|3\rangle}{3}.\end{aligned} \tag{33}$$

Accordingly, the eigenvalues of ρ^i corresponding to the eigenvector $|1\rangle$–$|3\rangle$ are all equal to $1/3$. Following (25), the quantum entropy $S(\rho^i)$ of the pixel matrices A and B is both 1.

Thus, for the composite quantum pixel matrix AB and its subsystems, and the quantum pixel matrices A and B, we can obtain

$$S\left(\rho_{AB}|\rho^i\right) = S(\rho_{AB}) - S\left(\rho^i\right) = -1 < 0. \tag{34}$$

The result of (34) implies that the quantum entropy of the composite quantum pixel matrix AB is smaller than the those of the quantum pixel matrices A and B. This conclusion is consistent with the fact that the properties of a composite quantum pixel matrix are determined by the individual quantum pixel matrices that constitute the matrix.

Moreover, since the trace of ρ_{AB} is equal to 1, it can be concluded that ρ_{AB} has a unique eigenvalue, that is, $\lambda = 1$. By combining this result with (22), the eigenvalues corresponding to ρ_{AB} can be solved by the characteristic equation, that is,

$$\left(\lambda E - \rho_{AB}\right)\chi = 0 \tag{35}$$

where E is the identity matrix and χ is the eigenvector of eigenvalue λ. By combining (22) with (35), we obtain

$$\left| E - \begin{bmatrix} a\bar{a} & a\bar{b} & a\bar{c} \\ b\bar{a} & b\bar{b} & b\bar{c} \\ c\bar{a} & c\bar{b} & c\bar{c} \end{bmatrix} \otimes \begin{bmatrix} x\bar{x} & x\bar{y} & x\bar{z} \\ y\bar{x} & y\bar{y} & y\bar{z} \\ z\bar{x} & z\bar{y} & z\bar{z} \end{bmatrix} \right| = 0. \tag{36}$$

By simplifying (36), we have

$$\begin{aligned}a^2\left(x^2+y^2+z^2\right) + b^2\left(x^2+y^2+z^2\right)\\ + c^2\left(x^2+y^2+z^2\right) = 1.\end{aligned} \tag{37}$$

Equation (37) can be verified by the normalization condition, which is satisfied by the eigenvectors corresponding to the quantum pixel matrices A and B, that is, $a^2+b^2+c^2 = 1$ and $x^2 + y^2 + z^2 = 1$. Therefore, the following relationship is established:

$$ayz = bcx. \tag{38}$$

Equation (38) specifies the condition for the entanglement of the quantum pixel matrices A and B. According to (3), the following equations are obtained:

$$\begin{aligned}a(x) &= \cos\left(\frac{G_R}{255}\right); \quad b(y) = \cos\left(\frac{G_G}{255}\right)\\ c(z) &= \cos\left(\frac{G_B}{255}\right).\end{aligned} \tag{39}$$

The quantum PME coefficient μ is used to describe the entanglement relationship between pixel matrices in this article, i.e.,

$$\mu = ayz - bcx. \tag{40}$$

With (39) and (40), when two quantum pixel matrices become entangled, the corresponding gray values exhibit the following relationship:

$$\begin{aligned}\mu = \cos\left(\frac{G_{R1}}{255}\right)\cos\left(\frac{G_{G2}}{255}\right)\cos\left(\frac{G_{B2}}{255}\right)\\ - \cos\left(\frac{G_{R2}}{255}\right)\cos\left(\frac{G_{G1}}{255}\right)\cos\left(\frac{G_{B1}}{255}\right).\end{aligned} \tag{41}$$

When the entanglement coefficient μ of the pixel matrix is zero, the corresponding gray values of the pixel matrices are equal, which can be expressed as

$$ayz = bcx; \quad a = x; \ b = y; \ c = z. \tag{42}$$

Combining the normalization conditions of the eigenvectors corresponding to the quantum pixel matrices A and B, i.e., $a^2+b^2+c^2 = 1$ and $x^2+y^2+z^2 = 1$, we obtained

$$a = x = b = y = c = z = \frac{\sqrt{3}}{3}. \tag{43}$$

In this case, the quantum pixel matrix AB, composed of quantum pixel matrices A and B, is in the maximally entangled state, and the PME occurs such that the corresponding entangled state reaches

$$|\phi_{AB}\rangle = \frac{\sqrt{3}}{3}(|11\rangle + |22\rangle + |33\rangle). \tag{44}$$

The process described above aims to extract the quantum matrix entanglement characteristics concealed at the pixel level of RSIs within the framework of quantum entanglement theory, namely, PME. First, the PME theory is developed to represent the quantum states of pixels using 3-D basis vectors in the Hilbert space, which can be expressed by (1) and (5). Second, the PME theory is proposed to quantify the corresponding entangled state of pixel matrices when the PME occurs, which can be expressed by (44). Together with the proposed PME theory, this article takes advantage of the rich spectral information, entanglement characteristics, and distance characteristics of pixel matrices in multispectral RSIs, as illustrated in Fig. 3.

C. PME Theory for Multispectral RSI Classification

The PME theory developed earlier may be applied to color image classification and further generalized to multispectral RSI classification. Let us thus consider a multispectral RSI with υ bands that has undergone relevant processing, such as noise reduction, geometric correction, and atmospheric correction. In particular, owing to the existence of composite pixel matrices composed of mixed pixels, an image segmentation method is employed to divide the unclassified image into several disjoint regions in this article. For each pixel of the image with $K_p = m \times n$ pixels, if $|\phi_i\rangle (i = 1, 2, \ldots, \upsilon)$ represents the eigenvector of the pixel in υ bands, then $|\phi\rangle$

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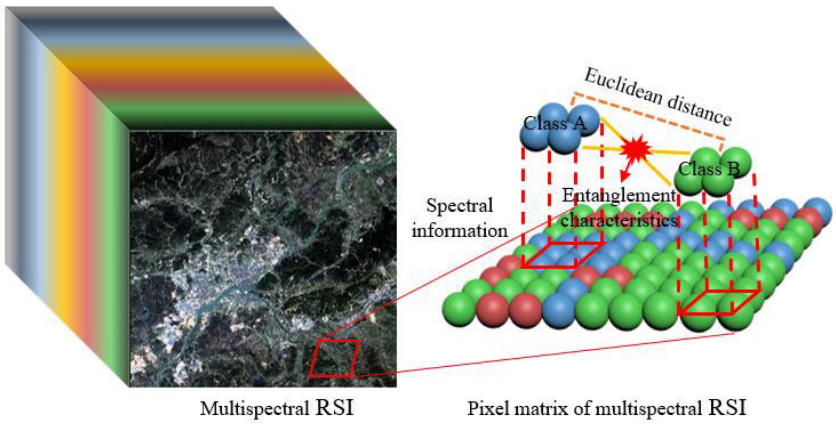


Fig. 3. Characteristics of PME.

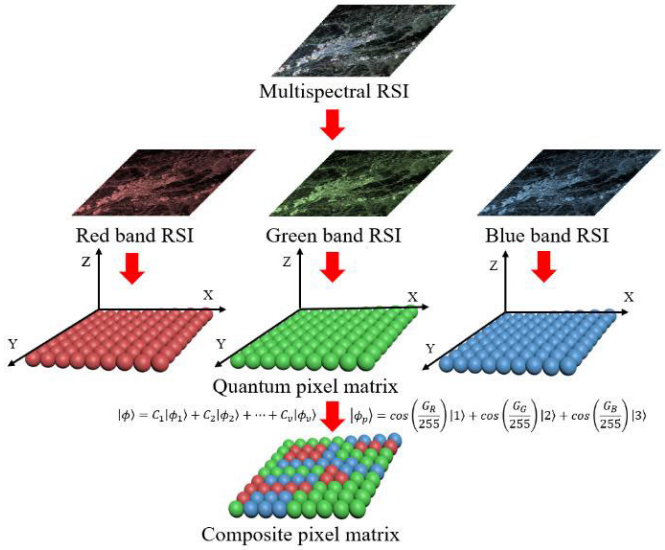


Fig. 4. Quantum pixel matrix of multispectral RSI.

contains its spectral information, and its quantum state can be defined by

$$|\phi\rangle = \sum_{i=1}^{\upsilon} C_i |\phi_i\rangle \tag{45}$$

where C_i is the eigenvalue of the eigenvector $|\phi_i\rangle$ with a value interval of [0, 1]; the definitions of C_i and $|\phi_i\rangle$ are

$$\begin{aligned} C_i &= \left[\cos\left(\frac{G_{Ri}}{255}\right), \cos\left(\frac{G_{Gi}}{255}\right), \cos\left(\frac{G_{Bi}}{255}\right)\right] \\ |\phi_i\rangle &= \alpha_i|1\rangle + \beta_i|2\rangle + \gamma|3\rangle. \end{aligned} \tag{46}$$

To fully illustrate the entanglement characteristics of the quantum pixel matrices at the pixel level of a multispectral RSI, the PME rule is proposed in this article. According to the PME theory proposed above, when it has three bands ($m \times n$), a multispectral RSI containing three bands can be represented as a composite pixel matrix with a 3-D orthogonal basis in the Hilbert space, as shown in Fig. 4.

The classification procedure begins by randomly selecting two pixel matrices, that is, pixel matrices C and D

$$\begin{aligned} |\phi_c\rangle &= |\phi_1\rangle + |\phi_2\rangle + |\phi_3\rangle \\ |\phi_D\rangle &= |\phi_a\rangle + |\phi_b\rangle + |\phi_c\rangle \end{aligned} \tag{47}$$

where $|\phi_1\rangle$–$|\phi_3\rangle$ are the quantum states of C and $|\phi_a\rangle$, $|\phi_b\rangle$, and $|\phi_c\rangle$ are the quantum states of D. Following (46), the quantum state of C is:

$$|\phi_c\rangle = \begin{bmatrix} \cos\left(\frac{G_{R1}}{255}\right) \\ \cos\left(\frac{G_{G1}}{255}\right) \\ \cos\left(\frac{G_{B1}}{255}\right) \end{bmatrix} + \begin{bmatrix} \cos\left(\frac{G_{R2}}{255}\right) \\ \cos\left(\frac{G_{G2}}{255}\right) \\ \cos\left(\frac{G_{B2}}{255}\right) \end{bmatrix} + \begin{bmatrix} \cos\left(\frac{G_{R3}}{255}\right) \\ \cos\left(\frac{G_{G3}}{255}\right) \\ \cos\left(\frac{G_{B3}}{255}\right) \end{bmatrix} \tag{48}$$

or

$$|\phi_c\rangle = \begin{bmatrix} a & b & c \end{bmatrix}^{\top} \tag{49}$$

where

$$\begin{aligned} a &= \cos\left(\frac{G_{R1}}{255}\right) + \cos\left(\frac{G_{R2}}{255}\right) + \cos\left(\frac{G_{R3}}{255}\right) \\ b &= \cos\left(\frac{G_{G1}}{255}\right) + \cos\left(\frac{G_{G2}}{255}\right) + \cos\left(\frac{G_{G3}}{255}\right) \\ c &= \cos\left(\frac{G_{B1}}{255}\right) + \cos\left(\frac{G_{B2}}{255}\right) + \cos\left(\frac{G_{B3}}{255}\right). \end{aligned} \tag{50}$$

Similarly, the quantum state of D is

$$|\phi_D\rangle = \begin{bmatrix} \cos\left(\frac{G_{Ra}}{255}\right) \\ \cos\left(\frac{G_{Ga}}{255}\right) \\ \cos\left(\frac{G_{Ba}}{255}\right) \end{bmatrix} + \begin{bmatrix} \cos\left(\frac{G_{Rb}}{255}\right) \\ \cos\left(\frac{G_{Gb}}{255}\right) \\ \cos\left(\frac{G_{Bb}}{255}\right) \end{bmatrix} + \begin{bmatrix} \cos\left(\frac{G_{Rc}}{255}\right) \\ \cos\left(\frac{G_{Gc}}{255}\right) \\ \cos\left(\frac{G_{Bc}}{255}\right) \end{bmatrix} \tag{51}$$

or

$$|\phi_D\rangle = \begin{bmatrix} x & y & z \end{bmatrix}^{\top} \tag{52}$$

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where

$$x = \cos\left(\frac{G_{Ra}}{255}\right) + \cos\left(\frac{G_{Rb}}{255}\right) + \cos\left(\frac{G_{Rc}}{255}\right)$$
$$y = \cos\left(\frac{G_{Ga}}{255}\right) + \cos\left(\frac{G_{Gb}}{255}\right) + \cos\left(\frac{G_{Gc}}{255}\right)$$
$$z = \cos\left(\frac{G_{Ba}}{255}\right) + \cos\left(\frac{G_{Bb}}{255}\right) + \cos\left(\frac{G_{Bc}}{255}\right). \quad (53)$$

According to the PME theory, the composite pixel matrix CD is in an entangled state when the pixel matrices C and D become entangled. Then, the PME coefficient μ can be used to describe the entanglement relationship between pixel matrices in the three-band multispectral image, i.e.,

$$\mu = ayz - bcx. \quad (54)$$

In the classification process of a multispectral image in this three-band image, when the quantum states $|\phi_1\rangle$–$|\phi_3\rangle$ of C and the quantum states $|\phi_a\rangle$, $|\phi_b\rangle$, and $|\phi_c\rangle$ of D reach the maximally entangled state, the entanglement coefficient μ of the composite CD equals 0, and the entangled state of the composite quantum matrix is a pure state with a quantum entropy of 0. If the entanglement coefficient μ of the composite CD is between 0 and 1, then the entangled state of the composite CD is between a mixed state and a pure state, which implies that the PME between C and D is occurring.

For multispectral image classification, the entanglement coefficient describes the degree of entanglement between different pixel matrices, related to the measurement probability of pixel matrices of different objects. Supposed that the entanglement coefficient μ of vegetation is 0, in this case, the measurement probability of pixel matrices reaches the maximum value, which implies that the measurement probability is lower when the entangled state of pixel matrices of vegetation is a mixed state or not in the maximally entangled state. The PME coefficient μ is proposed to extract the entanglement relationship between pixel matrices in the Hilbert space, which can be expressed by (54). Specifically, when the absolute value $|\mu|$ of entanglement coefficient μ is smaller, the entanglement degree between pixel matrices is actually larger, and the similarity between them is larger. In addition to the coefficient μ, the Euclidean distance d is also used to measure the similarity of pixel matrices.

Therefore, after computing the PME coefficient, one of the tasks of the ISOPME algorithm is to associate each quantum pixel of the image with the closest cluster center according to the Euclidean distance d and the PME coefficient μ, iteratively clustering the pixels, as shown in Fig. 5. In ISOPME classification criteria for RSI, the PME coefficient μ and the Euclidean distance d are used to measure the similarity of pixel matrices. If p_i is the eigenvector of the ith quantum pixel matrix in the above image, it can be inferred that

$$p_i = \left[p_{i(1)}, p_{i(2)}, \ldots, p_{i(K_P)}\right]^\top \quad (55)$$

where the eigenvectors of each quantum pixel are composed of three components, and the gray values of each component range from 0 to 255. If the image contains M types of ground objects, the proposed ISOPME algorithm must first select the initial cluster center K_C. The Euclidean distance d between the eigenvectors of a certain quantum pixel matrix and the cluster center is then computed as

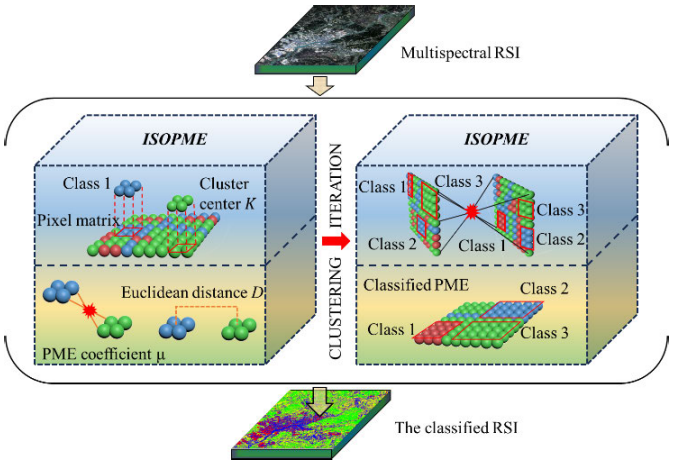


Fig. 5. PME theory for ISOPME classification.

$$d_{ij} = \sqrt{\sum_{a=1}^{K_p} \left(p_{i(a)} - K_{j(a)}\right)^2}, \quad (1 \ll j \ll M) \quad (56)$$

where $p_{i(a)}$ is the ath component of the ith eigenvector of the quantum pixel matrix and $K_{j(a)}$ is the ath component of the jth cluster center. K_p is the number of pixel matrices in the image. Once the minimum Euclidean distance (MED) d_{ij} is selected and the PME coefficient μ is as a clustering standard, the sample pixel matrix i will be associated with cluster j, indicating that they belong to the same category. In the next iteration, the new cluster center K'_j will be regenerated until the computed results converge and the image clustering results satisfy the classification requirements, K'_j defined as follows [57], [58], [59]:

$$K'_{j(a)} = \frac{1}{N_p} \sum_{i=1}^{N_p} p_{i(a)}(j), \quad (1 \ll a \ll K) \quad (57)$$

where N_p is the number of pixel matrices associated with the jth cluster and K is the number of cluster centers.

With the PME theory mentioned above for multispectral RSI classification, it can be analyzed that the PME coefficient μ and the Euclidean distance d together constitute the classification criteria used to measure the similarity of pixel matrices. Above all, the entanglement coefficient μ describes the similarity between pixel matrices by describing the entanglement degree between them. Specifically, when the absolute value $|\mu|$ of the entanglement coefficient μ is smaller, the entanglement degree between pixel matrices is actually larger, and the similarity between them is larger. If $|\mu|$ reaches the minimum value, i.e., $\mu = 0$, then the pixel matrices become entangled and reach the maximally entangled state. In this state, the eigenvalues of the entangled pixel matrices are equal to each other, which implies that their gray values are equal as well. Thus, the entangled pixel matrices are highly likely to be classified as the same object, given that they have the same spectral characteristics. However, due to the

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existence of the phenomenon known as "the same spectrum but different object," it is necessary to further determine the class of the object. The Euclidean distance d is therefore used to measure the similarity between pixel matrices after computing the entanglement coefficient μ. Also, when the Euclidean distance d is smaller, the similarity between pixel matrices is larger. Provided that d reaches its minimum value, the entangled pixel matrices are classified as the same object. Consequently, the proposed ISOPME method can effectively solve the phenomenon known as "the same object but different spectrum" as well.

D. Discussion of ISOPME Model

In this article, the mixed pixels in an image are considered as the quantum pixel matrices with quantum states, most of which are entangled. Therefore, the PME theory has been applied to extract the entanglement characteristics of quantum matrices concealed at the pixel level in an RSI, enabling the self-organizing clustering of images. The proposed ISOPME modelis developed to implement a self-organizing clustering for multispectral RSI. The detailed process of the ISOPME model is mainly composed of six parts, namely, pixel quantization, quantum pixel matrix computation, quantum state superposition of the multispectral image pixel matrices, initial cluster center setting, PME, and iterative self-organizing clustering, as shown in Fig. 6.

The ISOPME image classification algorithm fully exploits the relationship between the spectral information and the entanglement characteristics of the pixel matrices in multispectral RSI. More specifically, the following conditions hold.

1) When two or more pixel matrices become entangled and the entangled state of the subsystem forming the composite quantum pixel matrix is a mixed state, the corresponding quantum entropy is larger than 0. These pixel matrices are related to each other through PME, allowing the entanglement coefficient μ to reflect the linearly correlated relationship between their gray values.
2) When two or more pixel matrices become entangled and the entangled state of the composite quantum pixel matrix is a pure state, the corresponding quantum entropy is 0. When the quantum states of these pixel matrices in each image band are in the maximally entangled state, the entanglement coefficient μ of the composite quantum pixel matrix is equal to 0.

Using the PME theory developed above, we can analyze that the quantum states determine the properties of the corresponding quantum pixel matrices. More specifically, the following conditions hold.

1) The pixel matrix with the maximally entangled state (also known as the "Bell state") is considered a quantum entangled pixel matrix; at this stage, the corresponding entanglement coefficient μ is zero. The entangled state of the quantum entangled pixel matrix is a linear combination of the corresponding orthogonal eigenvectors. In other words, the gray values of these pixel matrices are linearly correlated.

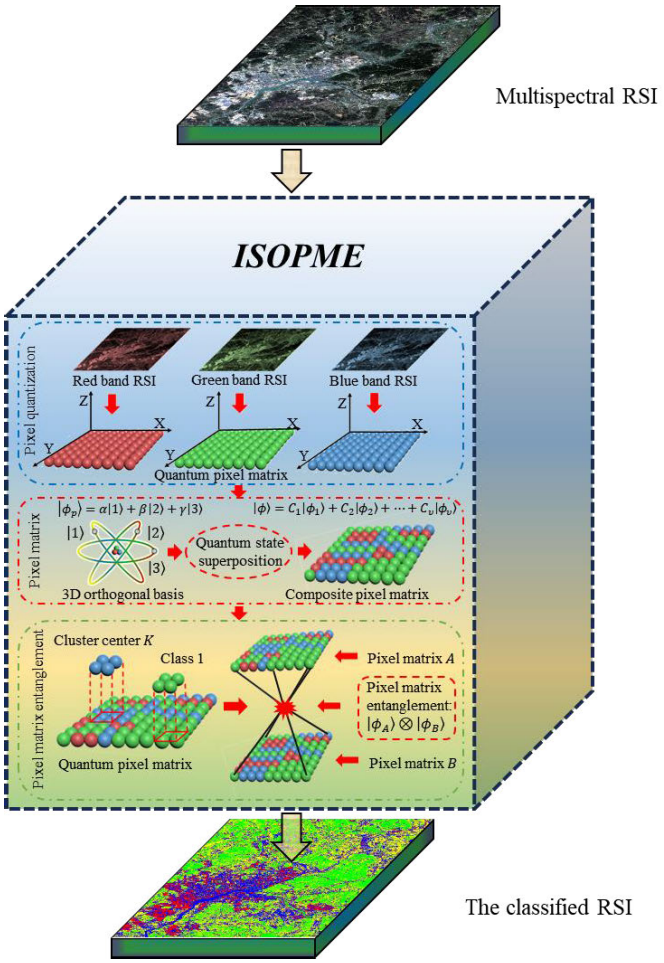


Fig. 6. ISOPME classification for RSI.

2) When the entanglement occurs between pixel matrices from the same RSI, the corresponding entanglement coefficient μ is zero. The entangled pixel matrices are classified as the same objects when the Euclidean distance d between them is minimal but nonzero, even if they are far apart or very close to each other, and the PME can still occur.

In summary, the major steps of the ISOPME method are given as follows.

1) Apply multispectral RSI preprocessing, including radiometric calibration, geometric correction, image cutting and fusion, image segmentation, and other steps.
2) Convert all the pixels of the RSI into quantum pixel matrices with a 3-D orthogonal basis in the Hilbert space.
3) Determine the number of bands in the image and manage the quantum state of each pixel matrix as a superposition of the corresponding quantum states in all bands.
4) Randomly select a few pixel matrices as the initial clusters and determine the center of each cluster.
5) Compute the entanglement coefficient μ between the pixel matrix for each object and the initial cluster centers.
6) Compute the Euclidean distance d between the pixel matrix for each object and the initial cluster centers.

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7) Perform splitting and merging operations to update the cluster centers.
8) Iterate points 5)–7) until the final converging status condition is reached.

The detailed steps of the ISOPME algorithm are shown in Algorithm 1.

Algorithm 1 ISOPME Algorithm

Step1: Multispectral RSI preprocessing.

Step2: Pixel quantization and construction of the quantum pixel matrices.

$$|\phi_p\rangle = \alpha|1\rangle + \beta|2\rangle + \gamma|3\rangle = \begin{bmatrix}\alpha & \beta & \lambda\end{bmatrix}^{\top};$$
$$\alpha = \cos\left(\frac{G_R}{255}\right); \beta = \cos\left(\frac{G_G}{255}\right); \gamma = \cos\left(\frac{G_B}{255}\right).$$

Step3: Quantum state superposition of multispectral image pixels.

$$|\phi\rangle = C_1|\phi_1\rangle + C_2|\phi_2\rangle + \cdots + C_v|\phi_v\rangle$$

Step4: Data input, input image and categories of ground object.

Step5: Initial cluster center K_C setting.

Step6: Pixel matrix entanglement coefficient μ computation, compute the entanglement coefficient μ between the quantum pixel matrix for each object and the initial cluster center K_C.

$$\mu = ayz - bcx.$$

Step7: Euclidean distance d computation, compute the Euclidean distance d between the pixel matrix for each object and initial cluster center K_C.

$$d_{ij} = \sqrt{\sum \left(p_{i(a)} - K_{j(a)}\right)^2}, (1 \ll j \ll M).$$

Step8: Conduct iteration based on the minimum Euclidean distance d' and the pixel matrix entanglement coefficient μ.

Step9: Automatically perform split and merge operations to update cluster center K'_j.

$$K'_{j(a)} = \frac{1}{N_p}\sum_{i=1}^{N_p} p_{i(a)}(j), (1 \ll a \ll K).$$

Step10: Iterate until iteration convergence.

Step11: End classification until the results satisfy the classification requirements.

Step12: Data output, output the results of classification.

III. Experiments

Fig. 7 shows a flowchart of the proposed ISOPME classification experiment, which is composed of four parts: image preprocessing, quantum pixel matrix computation, PME, and iterative self-organizing clustering.

A. Four Test Areas

Four test areas were selected to validate the proposed ISOPME model, as illustrated in Fig. 8. The dataset has been a public dataset, as provided by the Geospatial Data Cloud site, Computer Network Information Center, Chinese Academy of Sciences, at http://www.gscloud.cn. The first test area is a coastal urban area characterized by clustered habitations, flat terrain, and dense vegetation. The second test area is a hilly area characterized by abundant crops, continuous hills, and rugged terrain. The third test area is a desert area characterized by infrequent rainfall, scattered populations, and extensive desertification. The fourth test area is a plateau area characterized by high altitude, rare vegetation, and rugged mountains. The image classification results of ISOPME are compared with those of ISODATA, K-Means, SOM, FCM, SVM, MDC, and DL models. The TensorFlow model in the DL method used in this article is trained by Inception-v3 Net, a deep neural network (DNN) model [60].

This process of TensorFlow model is depicted in Fig. 9, which is composed of four parts.

1) RSI input, used to import the unclassified multispectral image data, including training and test data.
2) TensorFlow preprocessing, which is for converting the images into pixel matrix data flow and Tensor form.
3) TensorFlow model training, which mainly consists of three sublayers: the input layer, hidden layer, and output layer. The input layer is for receiving the TensorFlow preprocessing data and initializing the weights and biases of the model, the hidden layer is to update the weights and iteratively to extract RSI spatial and spectral features based on the marked samples, and the output layer is designed for the final classification of the images until the iteration converges.
4) Classification output, whose functionality is to export the classified images.

1) First Test Area: The first test area is located in Beihai City, Guangxi Zhuang Autonomous Region, China, from 108°59′E to 109°25′E and from 21°20′N to 21°44′N. An October 2021 Landsat-8 image with a total of 11 bands is considered, including four visible light bands, three short-wave infrared bands, two thermal infrared bands, one near-infrared band, and one panchromatic band, for an actual area of approximately 2061 km^2, which contains roughly 20% habitations (HNs), 35% vegetation (VN), 34% water (WR), and 9% soil (SL). The Landsat-8 satellite, launched on February 11, 2013, is on a near-polar Sun-synchronous orbit with an orbital altitude of 705 km and carries two sensors: the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). The image has a spatial resolution of 30 m in the multispectral and thermal infrared bands and 15 m in the panchromatic band, and the temporal resolution of the image is 16 days.

2) Second Test Area: The second test area is located in Shangrao City, Jiangxi Province, China, from 117°48′E to 118°10′E and from 28°18′N to 28°37′N. A July 2021 Landsat 8 image with a total of 11 bands is considered, for an actual area of approximately 1218 km^2, which contains roughly 15% HNs, 33% VN, 24% WR, and 35% SL.

3) Third Test Area: The third test area is located in Ordos City, Inner Mongolia Autonomous Region, China, from 108°28′E to 108°46′E and from 40°37′N to 40°50′N. An August 2021 Landsat 8 image with a total of 11 bands is considered, for an actual area of approximately 632 km^2, which contains roughly 25% HNs, 18% VN, 30% WR, and 24% SL.

4) Fourth Test Area: The fourth test area is located in Xining City, Qinghai Province, China, from 101°36′E to

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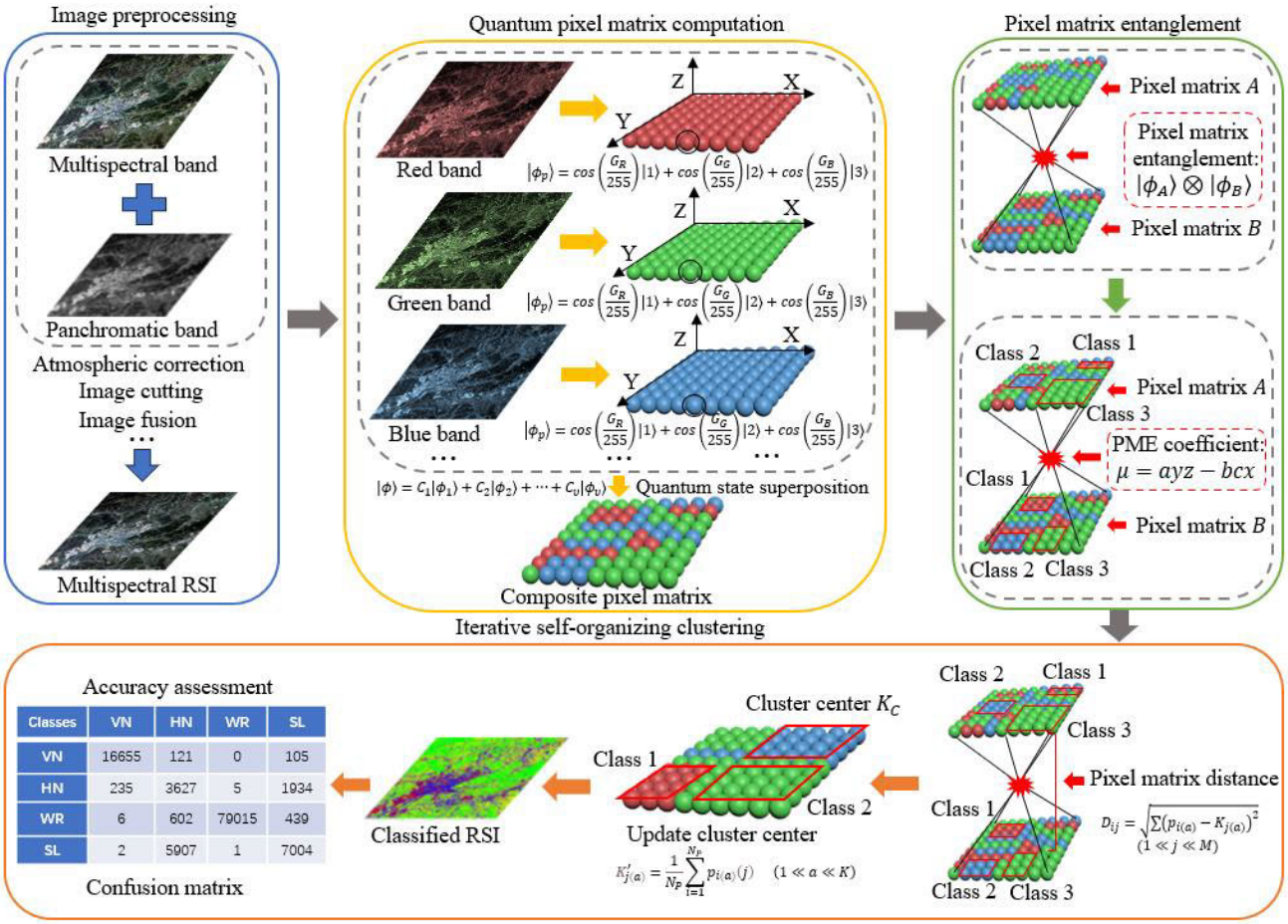


Fig. 7. Flowchart of ISOPME classification experiment.

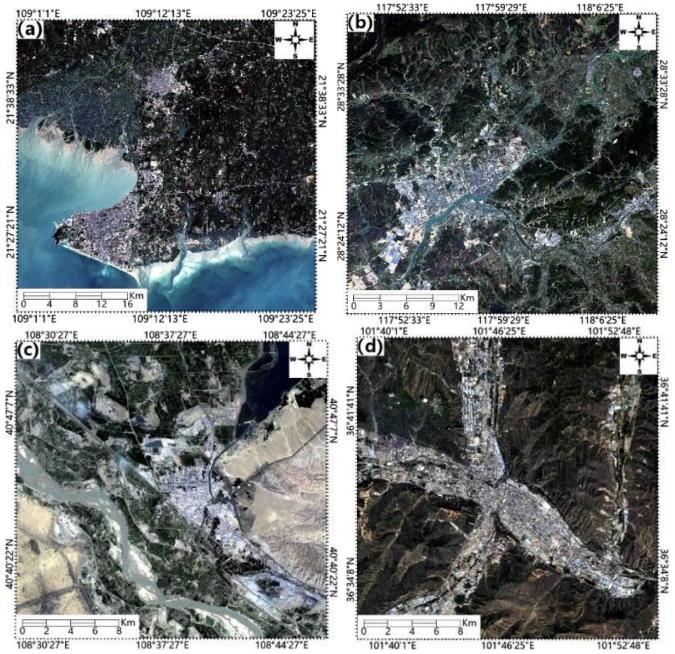


Fig. 8. (a) First test area. (b) Second test area. (c) Third test area. (d) Fourth test area.

101°54′E and from 36°31′N to 36°46′N. An August 2020 Landsat 8 image with a total of 11 bands is considered, for an actual area of approximately 620 km^2, which contains roughly 18% HNs, 42% VN, and 36% SL.

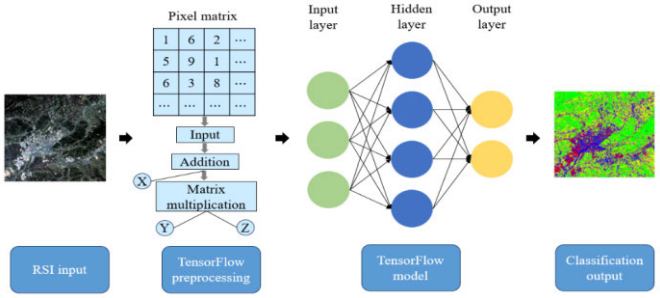


Fig. 9. Workflow of the TensorFlow model in DL classification.

B. Verification Experiments in Test Areas

1) Verification in the First Test Area: The first test area was classified into four classes: HN, VN, WR, and SL. The detailed classification process is given as follows.

1) *Multispectral RSI Preprocessing:* It includes radiometric calibration, geometric correction, image cutting and fusion, image segmentation, and other related steps.
2) *Quantum Pixel Matrix Computation:* Convert all pixels of the RSI into quantum pixel matrices with a 3-D

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orthogonal basis in the Hilbert space

$$|\phi_p\rangle = \alpha|1\rangle + \beta|2\rangle + \gamma|3\rangle = \begin{bmatrix}\alpha & \beta & \gamma\end{bmatrix}^\top$$
$$\alpha = \cos\left(\frac{G_R}{255}\right);\ \beta = \cos\left(\frac{G_G}{255}\right);\ \gamma = \cos\left(\frac{G_B}{255}\right). \tag{58}$$

3) *Quantum State Superposition of the Pixel Matrices:* Determine the number of bands, υ, and cope with the quantum state $|\phi\rangle$ of each pixel matrix as a superposition of corresponding quantum states in all υ band images. The image in the first test area has 11 bands ($\upsilon = 11$)

$$|\phi\rangle = C_1|\phi_1\rangle + C_2|\phi_2\rangle + \cdots + C_\upsilon|\phi_\upsilon\rangle$$
$$C_\upsilon = \left[\cos\left(\frac{G_{R\upsilon}}{255}\right) \quad \cos\left(\frac{G_{G\upsilon}}{255}\right) \quad \cos\left(\frac{G_{B\upsilon}}{255}\right)\right]. \tag{59}$$

4) *Data Input:* Input the unclassified image and categories of ground object ($M = 4$).
5) *Initial Cluster Center K_C Setting:* Randomly select a few pixel matrices as the initial clusters and determine the center K_C of each cluster whose number is K.
6) *PME:* Compute the entanglement coefficient μ between the quantum pixel matrix for each object and initial cluster center K_C

$$|\phi_c\rangle = a|1\rangle + b|2\rangle + c|3\rangle$$
$$|\phi_D\rangle = x|1\rangle + y|2\rangle + z|3\rangle$$
$$\mu = a\text{yz-b}cx \tag{60}$$

where a, b, and c are the eigenvalues of pixel matrix C; and x, y, and z are the eigenvalues of pixel matrix D.
7) *Iterative Self-Organizing Clustering:* Compute the Euclidean distance d between the pixel matrix for each category and the initial cluster center K_C, and conduct the iteration operation based on the MED d' and the PME coefficient μ; then, automatically perform split and merge operations to update the cluster center K'_j

$$d_{ij} = \sqrt{\sum_{i=1}^{K_p}\left(p_{i(a)} - K_{j(a)}\right)^2}, \quad (1 \ll j \le 4)$$
$$K'_{j(a)} = \frac{1}{N_p}\sum_{i=1}^{N_p} p_{i(a)}(j), \quad (1 \ll a \ll 4) \tag{61}$$

where $p_{i(a)}$ is the ath component of the ith eigenvector of the quantum pixel matrix, $K_{j(a)}$ is the ath component of the jth cluster center, and N_p is the number of pixel matrices associated with cluster j in the unclassified image.
8) *Data Output:* Repeat the iterative process above until the given converge condition is satisfied and output the results of image classification that meet the requirements.
9) *Comparative Analysis:* To verify the proposed ISOPME method, the ISODATA, K-Means, SOM, FCM, SVM, MDC, and DL methods are selected for comparative analysis. The producer accuracy (PA) and user accuracy (UA) are employed to evaluate the classification performance of each method. The final results for the first test area are shown in Fig. 10, and the confusion matrices for the classification results are shown in Table I.

According to the confusion matrices of the classification results in Table I and the PA and UA of the classification results in the first test area, the proposed ISOPME method has higher classification accuracy for HN, VN, and WR but lower classification accuracy for SL than the four unsupervised classification methods (ISODATA, K-Means, SOM, and FCM); it has higher classification accuracy for WR and SL but lower classification accuracy for HN and VN than the two supervised classification methods (SVM and MDC); it has higher classification accuracy for WR and SL but lower classification accuracy for HN and VN than the DL method; it has higher average PA and UA than the four unsupervised classification methods, with an average increase of 9.84% and 14.97%, respectively, but lower average PA and UA than the SVM and DL methods.

2) Verification in the Second Test Area: The second test area is classified into four classes: HN, VN, WR, and SL. As shown in the classification process of the first test area, the detailed classification process of the second test area is also composed of nine parts, namely, multispectral RSIs preprocessing, quantum pixel matrix computation, quantum state superposition of the multispectral image pixel matrices, data input, initial cluster center K_C setting, PME, iterative self-organizing clustering, data output, and comparative analysis. The final results for the second test area are shown in Fig. 11 and the confusion matrices for the classification results are shown in Table II.

According to the confusion matrix of the classification results in Table II and the PA and UA of the classification results in the second test area, the proposed ISOPME method has higher classification accuracy for VN, WR, and SL but lower classification accuracy for HN than the four unsupervised classification methods; it has higher classification accuracy for VN and SL categories but lower classification accuracy for HN and WR than the supervised classification methods; it has higher classification accuracy for VN and SL but lower classification accuracy for HN and WR than the DL method; and it has higher average PA and UA than the four unsupervised classification methods, with an average increase of 11.98% and 11.65%, respectively, and reaches the same accuracy level when compared with the SVM, MDC, and DL methods.

3) Verification in the Third Test Area: The third test area is classified into four classes: HN, VN, WR, and SL. As shown in the classification process of the first test area, the detailed classification of the third test area is also composed of nine parts, namely, multispectral RSIs preprocessing, quantum pixel matrix computation, quantum state superposition of the multispectral image pixel matrices, data input, initial cluster center K_C setting, PME, iterative self-organizing clustering, data output, and comparative analysis. The results for the third test area are shown in Fig. 12 and the confusion matrices of the classification results are shown in Table III.

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TABLE I

CONFUSION MATRICES FOR THE FIRST TEST AREA

Methods	Classes	VN	HN	WR	SL	PA	UA	Methods	Classes	VN	HN	WR	SL	PA	UA
ISOPME	VN	16655	121	0	105	98.56%	98.66%	ISODATA	VN	16592	18	0	33	98.19%	99.69%
	HN	235	3627	5	1934	35.36%	62.52%		HN	305	1939	1391	1759	18.90%	35.95%
	WR	6	602	79015	439	99.99%	98.69%		WR	0	1018	77639	422	98.24%	98.18%
	SL	2	5907	1	7004	73.87%	54.24%		SL	1	7282	0	7268	76.65%	49.95%
K-Means	VN	16592	18	0	33	98.19%	99.69%	SOM	VN	15792	53	0	51	93.45%	99.35%
	HN	305	1939	1391	1759	18.90%	35.95%		HN	1095	1562	3331	7332	15.23%	23.24%
	WR	0	1018	77639	422	98.24%	98.18%		WR	6	2846	75690	1339	95.78%	94.75%
	SL	1	7282	0	7268	76.65%	49.95%		SL	5	5796	0	7360	77.62%	55.92%
FCM	VN	15792	53	0	51	93.45%	99.35%	SVM	VN	17869	163	0	113	99.66%	98.48%
	HN	1095	1562	22549	732	15.23%	6.02%		HN	49	10685	0	4061	89.30%	72.22%
	WR	6	2846	56470	1339	71.46%	93.09%		WR	8	194	79015	8	96.07%	99.73%
	SL	5	5796	2	7360	77.62%	55.91%		SL	4	923	3231	6586	61.16%	61.30%
MDC	VN	17917	1841	0	2141	99.93%	81.82%	DL	VN	17917	184	0	117	99.93%	98.35%
	HN	3	4897	0	1579	40.93%	75.58%		HN	12	11302	0	4063	94.46%	73.50%
	WR	9	1481	79020	188	96.08%	97.92%		WR	0	156	79017	9	96.07%	99.79%
	SL	1	3746	3226	6860	63.71%	49.59%		SL	1	323	3229	6579	61.10%	64.93%

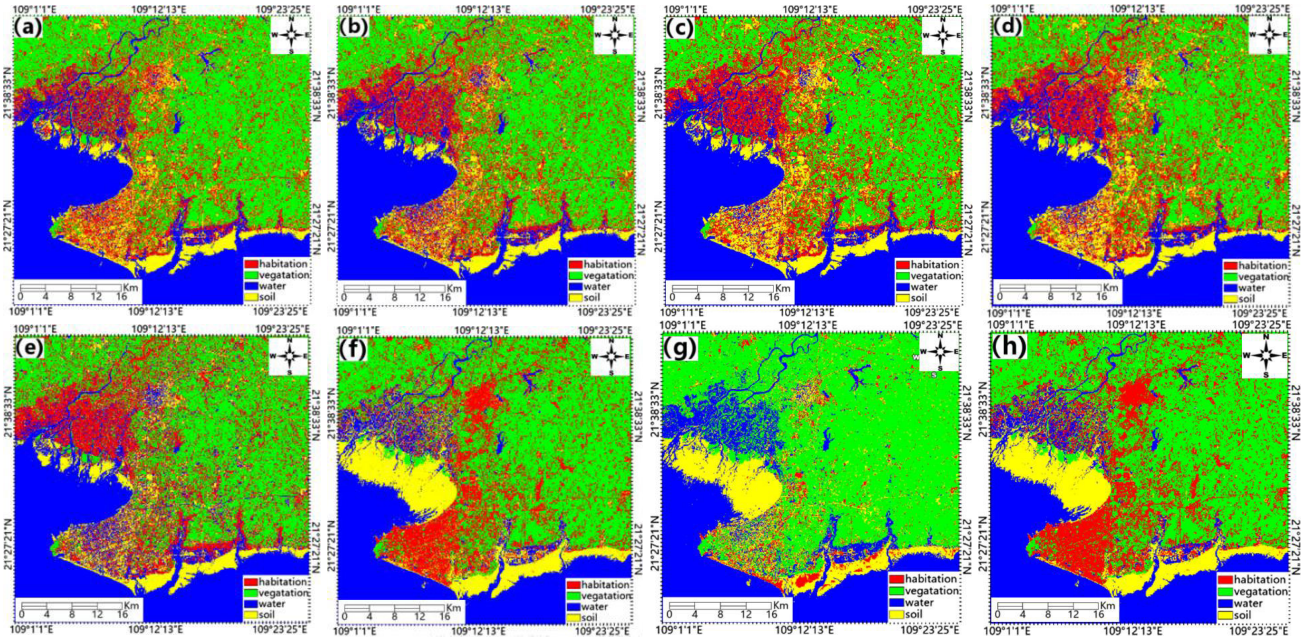


Fig. 10. Classification results of the first test area from (a) ISOPME method, (b) ISODATA method, (c) K-Means method, (d) SOM method, (e) FCM method, (f) SVM method, (g) MDC method, and (h) DL method.

TABLE II

CONFUSION MATRICES FOR THE SECOND TEST AREA

Methods	Classes	VN	HN	WR	SL	PA	UA	Methods	Classes	VN	HN	WR	SL	PA	UA
ISOPME	VN	6406	5	987	71	97.40%	85.77%	ISODATA	VN	6077	0	475	0	92.40%	92.75%
	HN	0	6335	1	3	90.17%	99.94%		HN	0	6907	1	3	98.31%	99.94%
	WR	1	622	5598	219	84.96%	86.93%		WR	4	116	5560	1473	84.38%	77.73%
	SL	170	64	3	4707	94.14%	95.21%		SL	496	3	553	3524	70.48%	77.01%
K-Means	VN	6077	0	475	0	92.40%	92.75%	SOM	VN	5868	1	391	8	89.22%	93.62%
	HN	0	6907	1	3	98.31%	99.94%		HN	1	6797	5	23	96.74%	99.58%
	WR	4	116	5560	1473	84.38%	77.73%		WR	3	179	3334	1234	50.60%	70.19%
	SL	496	3	553	3524	70.48%	77.01%		SL	705	49	2859	3735	74.70%	50.83%
FCM	VN	5786	0	348	0	87.97%	94.33%	SVM	VN	6170	65	1	87	96.53%	97.58%
	HN	1	6964	4	16	99.12%	99.70%		HN	1	7287	10	1711	95.81%	80.89%
	WR	2	58	3424	1482	51.97%	68.95%		WR	0	85	6577	13	99.82%	98.53%
	SL	788	4	2813	3502	70.04%	49.28%		SL	221	169	1	4930	73.13%	92.65%
MDC	VN	6157	96	17	98	96.32%	96.69%	DL	VN	6266	111	1	108	98.03%	96.61%
	HN	0	6550	1	1282	86.12%	83.62%		HN	1	7317	11	1719	96.20%	80.87%
	WR	0	438	6570	17	99.71%	93.52%		WR	0	106	6577	13	99.82%	98.22%
	SL	235	522	1	5344	79.28%	87.58%		SL	125	72	0	4901	72.70%	96.14%

According to the confusion matrix of the classification results in Table III and the PA and UA of the classification results in the third test area, the proposed ISOPME method has higher classification accuracy for HN, WR, and SL but lower classification accuracy for VN than the four unsupervised classification methods; it has higher

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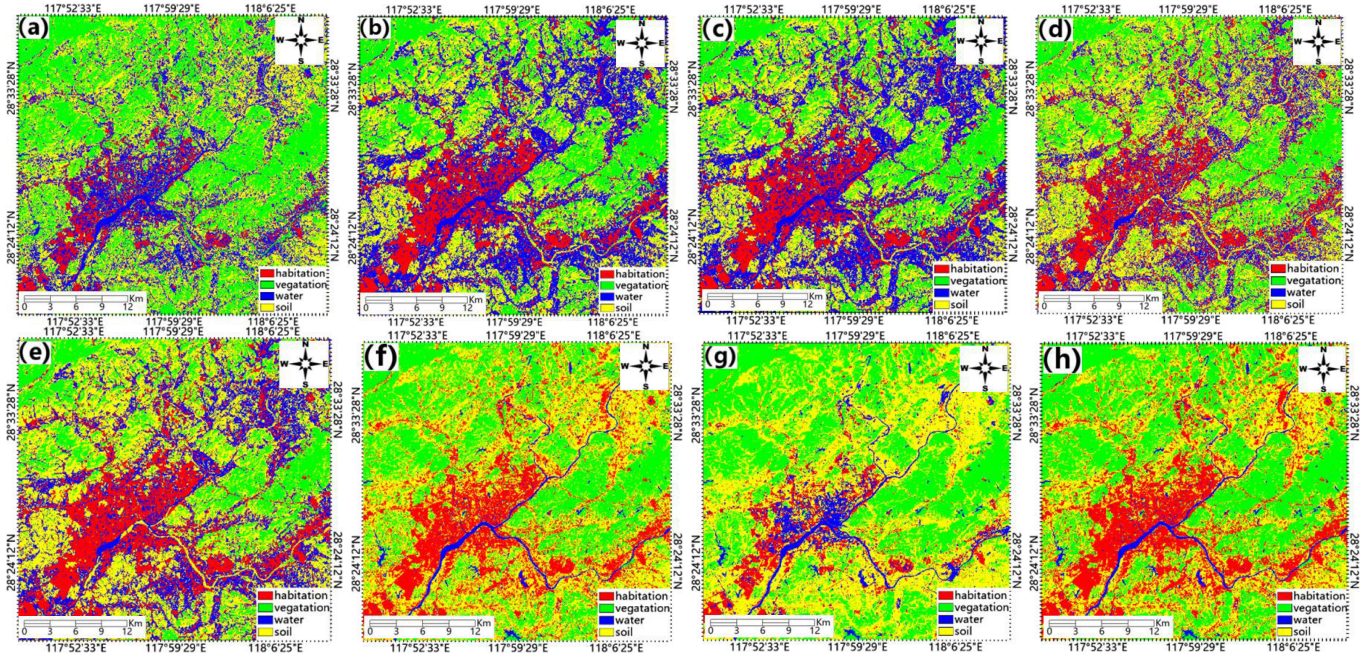


Fig. 11. Classification results of the second test area from (a) ISOPME method, (b) ISODATA method, (c) K-Means method, (d) SOM method, (e) FCM method, (f) SVM method, (g) MDC method, and (h) DL method.

TABLE III
CONFUSION MATRICES FOR THE THIRD TEST AREA

Methods	Classes	VN	HN	WR	SL	PA	UA	Methods	Classes	VN	HN	WR	SL	PA	UA
ISOPME	VN	3875	0	782	0	92.11%	83.21%	ISODATA	VN	3972	1	782	54	99.00%	82.60%
	HN	0	1931	0	0	96.55%	100.00%		HN	0	1866	0	0	93.30%	100.00%
	WR	129	68	1789	2	69.58%	89.99%		WR	1	116	14	113	0.72%	5.74%
	SL	203	1	0	1637	99.88%	88.92%		SL	39	17	1144	1472	89.81%	55.09%
K-Means	VN	3972	1	782	54	99.00%	82.60%	SOM	VN	3971	0	1396	45	98.98%	73.37%
	HN	0	1866	0	0	93.30%	100.00%		HN	0	1845	0	0	92.25%	100.00%
	WR	1	116	14	113	0.72%	5.74%		WR	1	144	1158	42	45.04%	86.10%
	SL	39	17	1144	1472	89.81%	55.09%		SL	40	11	17	1152	94.69%	95.80%
FCM	VN	3972	1	1492	54	99.00%	71.97%	SVM	VN	3964	0	0	13	98.80%	99.67%
	HN	0	1866	0	0	93.30%	100.00%		HN	0	2175	0	948	98.24%	69.64%
	WR	1	116	14	113	0.53%	5.74%		WR	2	0	2650	0	100.00%	99.92%
	SL	39	17	1144	1472	89.81%	55.09%		SL	46	39	0	1632	62.94%	95.05%
MDC	VN	3950	0	0	134	93.89%	96.72%	DL	VN	3968	0	0	13	94.32%	99.67%
	HN	0	1917	0	409	86.59%	82.42%		HN	13	2175	0	715	98.24%	74.92%
	WR	39	1	2650	3	100.00%	98.40%		WR	2	0	2650	0	100.00%	99.92%
	SL	218	296	0	1809	76.82%	77.87%		SL	224	39	0	1627	69.09%	86.08%

classification accuracy for SL but lower classification accuracy for VN, HN, and WR than the supervised classification methods; it has higher classification accuracy for SL but lower classification accuracy for VN, HN, and WR than the DL method; and it has higher average PA and UA than the four unsupervised classification methods, with an average increase of 9.84% and 14.97%, respectively, and reaches the same accuracy level when compared with the SVM, MDC, and DL methods.

4) Verification in the Fourth Test Area: The fourth test area was classified into three classes: HN, VN, and SL. As shown in the classification process of the first test area, the detailed classification process of the fourth test area is also composed of nine parts, namely, multispectral RSIs preprocessing, quantum pixel matrix computation, quantum state superposition of the multispectral image pixel matrices, data input, initial cluster center K_C setting, PME, iterative self-organizing clustering, data output, and comparative analysis. The classification results are shown in Fig. 13 and the confusion matrices of the classification results are shown in Table IV.

According to the confusion matrix of the classification results in Table IV and the PA and UA of the classification results in the fourth test area, the proposed ISOPME method has higher classification accuracy for VN and SL but lower classification accuracy for HN than the unsupervised classification methods; it has higher classification accuracy for HN but lower classification accuracy for VN and SL than the two supervised classification methods; it has higher classification accuracy for HN but lower classification accuracy for VN and

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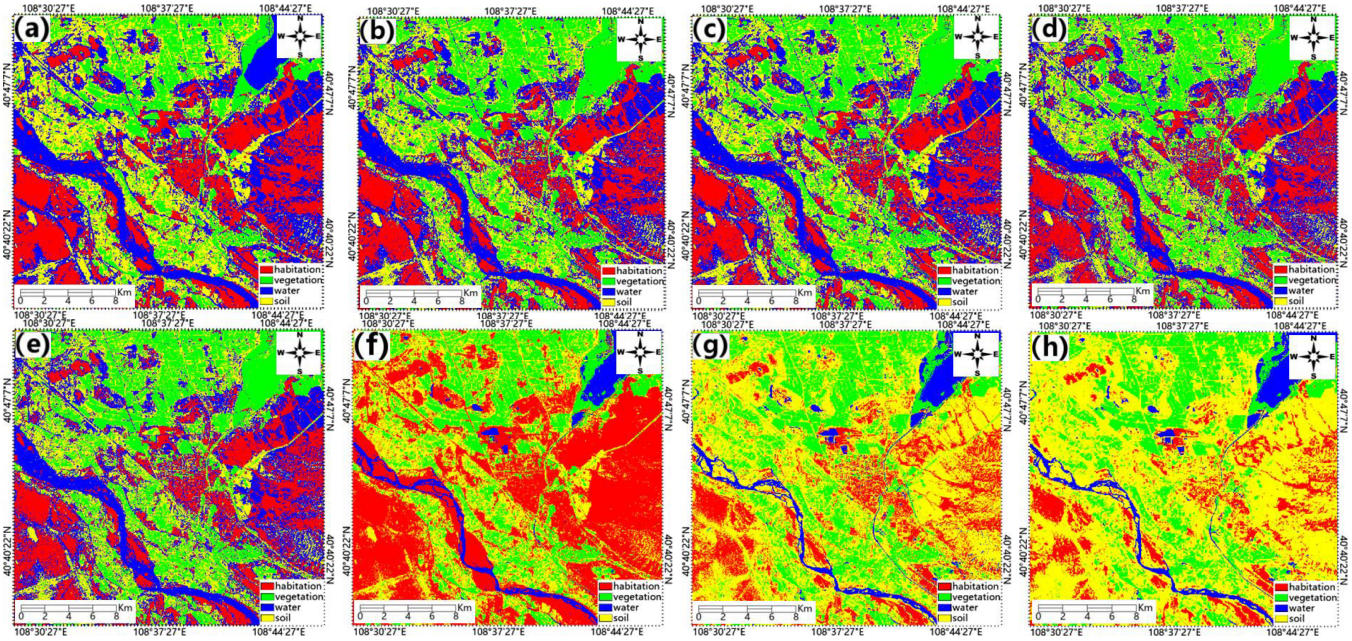


Fig. 12. Classification results of the third test area from (a) ISOPME method, (b) ISODATA method, (c) K-Means method, (d) SOM method, (e) FCM method, (f) SVM method, (g) MDC method, and (h) DL method.

TABLE IV
CONFUSION MATRICES FOR THE FOURTH TEST AREA

Methods	Classes	VN	HN	SL	PA	UA	Methods	Classes	VN	HN	SL	PA	UA
ISOPME	VN	5626	3	14	99.33%	99.70%	ISODATA	VN	5088	2	3	89.83%	99.90%
	HN	0	4076	396	99.34%	91.14%		HN	0	4066	669	99.10%	85.87%
	SL	38	24	1947	82.61%	96.91%		SL	576	35	1685	71.49%	73.39%
K-Means	VN	5088	2	3	89.83%	99.90%	SOM	VN	4979	0	0	87.91%	100.00%
	HN	0	4066	669	99.10%	85.87%		HN	0	4097	767	99.85%	84.23%
	SL	576	35	1685	71.49%	73.39%		SL	685	6	1590	67.46%	69.71%
FCM	VN	4937	0	0	87.16%	100.00%	SVM	VN	5628	5	17	99.36%	99.61%
	HN	0	4096	778	99.83%	84.04%		HN	0	3871	32	94.35%	99.18%
	SL	727	7	1579	66.99%	68.27%		SL	36	227	2308	97.92%	89.77%
MDC	VN	5647	5	51	99.70%	99.02%	DL	VN	5653	5	19	99.81%	99.58%
	HN	0	3115	41	75.92%	98.70%		HN	0	3950	34	96.27%	99.15%
	SL	17	983	2265	96.10%	69.37%		SL	11	148	2304	97.75%	93.54%

SL than the DL method; and it has higher PA and UA than the four unsupervised classification, with an average increase of 11.98% and 11.65%, respectively, but lower average PA and UA than the SVM and DL methods.

The proposed ISOPME method works better because our method combines several properties of pixels, including the spectrum, PME, and Euclidean distance, plus the use of a new measurement index and the similarity of the object pixels. Not all of these properties have been fully applied in the early unsupervised and supervised classification models. For example, the measurement of the similarity for the VN pixels using the proposed ISOPME is larger than that from ISODATA, which demonstrates that the proposed ISOPME works better for the VN class than ISODATA. On the other hand, the proposed ISOPME sometimes works worse than other supervised models that our method is an unsupervised method. This implies that no prior knowledge and no training samples are used, resulting in a low classification accuracy than the supervised classification, such as SVM, which can be demonstrated by the classification results in the first test area.

IV. RESULTS AND DISCUSSION

A statistical analysis of the proportion of classes for all methods is shown in Fig. 15 to evaluate the classification accuracy of the ISOPME model.

From the 3-D stacked percentage bar chart in Fig. 15, it can be analyzed that: 1) the ISOPME model has a similar proportion of objects in classification results compared to ISODATA, but it demonstrates higher classification accuracy when combined with the overall accuracy (OA) and Kappa coefficient (KC) in Table V; 2) in the first test area, the

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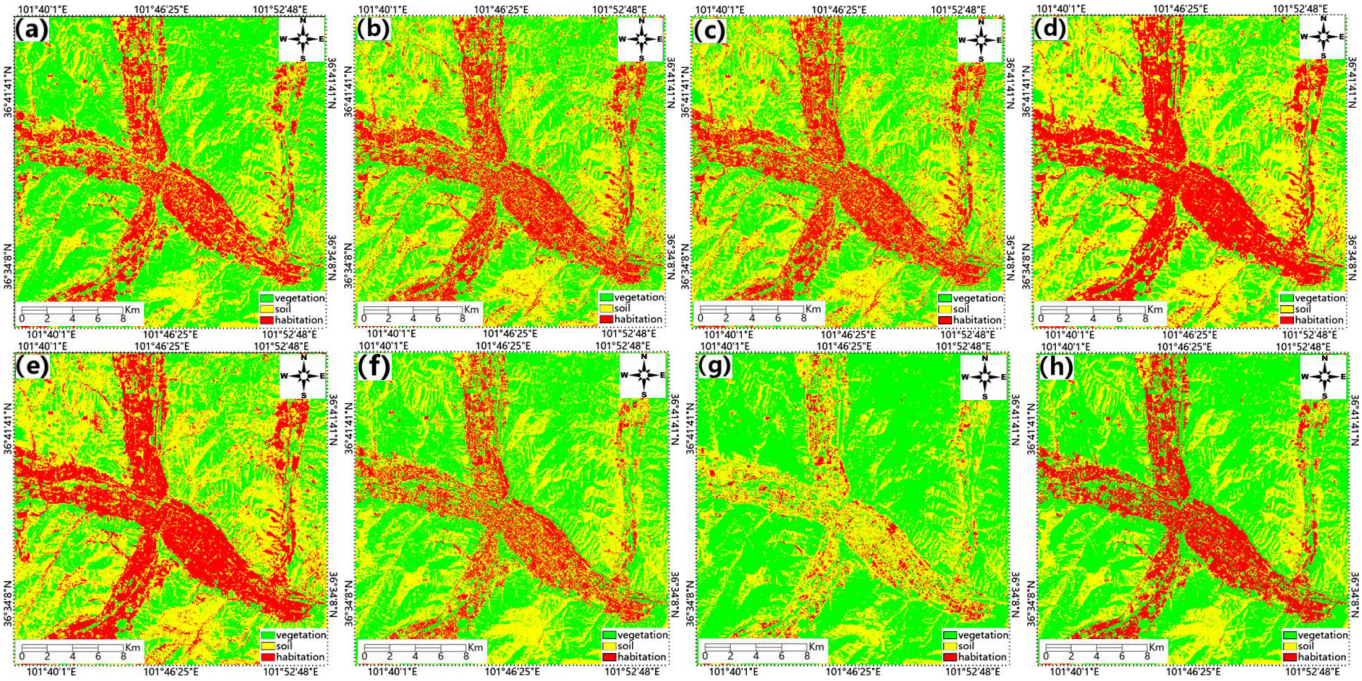


Fig. 13. Classification results of the fourth test area from (a) ISOPME method, (b) ISODATA method, (c) K-Means method, (d) SOM method, (e) FCM method, (f) SVM method, (g) MDC method, and (h) DL method.

ISOPME model has a higher proportion of VN than the four unsupervised classification models; it has a lower proportion of SL and VN than the supervised classification models; it has a lower proportion of SL and VN than the DL model; 3) in the second test area, the ISOPME model has a higher proportion of VN than the four unsupervised classification models; it has a lower proportion of SL and HN than the supervised classification models; it has a lower proportion of SL, HN, and VN than the DL model; 4) in the third test area, the ISOPME model has a higher proportion of SL and WR than the four unsupervised classification models; it has a lower proportion of SL and VN than the supervised classification models; it has a lower proportion of SL and VN than the DL model; and 5) in the fourth test area, the ISOPME model has a higher proportion of VN than the four unsupervised classification models; it has a lower proportion of SL than the supervised classification models; and it has a lower proportion of VN than the DL model.

A. Comparison of Classification Accuracy

In order to further compare the classification accuracy of the proposed ISOPME model with other classification models, this article employs the confusion matrices for classification results from four test areas and calculates the corresponding accuracy evaluation indicators, including the OA, and KC. The confusion matrices and accuracy evaluation indicators are detailed in Tables I–V to verify the accuracy of the predictive and the true values of the categories of each measured pixel in the RSI classification results.

According to the classification results of four test areas mentioned above, the following conclusions can be drawn: 1) the proposed ISOPME has the highest classification accuracy among five unsupervised classification methods; the OA of ISOPME increases by an average of 8.96% and the KC increases by an average of 0.14 in four test areas compared with the four traditional unsupervised classification models; 2) the classification accuracy of ISOPME is commensurate to the supervised classification models; the OA of ISOPME decreases by an average of 1.10%, and the KC decreases by an average of 0.02 in four test areas compared with the traditional supervised classification model of SVM, and however, the SVM model requires more prior knowledge; and 3) the classification accuracy of ISOPME is close to that of the DL model; the OA of ISOPME decreases by an average of 1.60%, and the KC decreases by an average of 0.03 in four test areas compared with the DL model; and however, the DL model requires more training samples.

The experimental results demonstrate that the proposed ISOPME model based on PME theory achieves higher classification accuracy than other unsupervised classification models. This is because the proposed ISOPME model makes full use of the features of pixel matrices, including the spectral information, the quantum entanglement characteristics, and the Euclidean distances between pixel matrices, while the traditional unsupervised classification methods, such as ISODATA, K-Means, SOM, and FCM, only consider the Euclidean distances between pixels, resulting in low classification accuracy. In the classification experiments of four test areas, the OA of ISOPME proposed in this article improves by 2.77%, 4.43%, 16.06%, and 7.47% compared to ISODATA and K-Means. Compared to SOM, it improves by 5.87%, 16.79%, 6.25%, and 9.22%. Compared to FCM, it improves by 30.94%, 17.13%,

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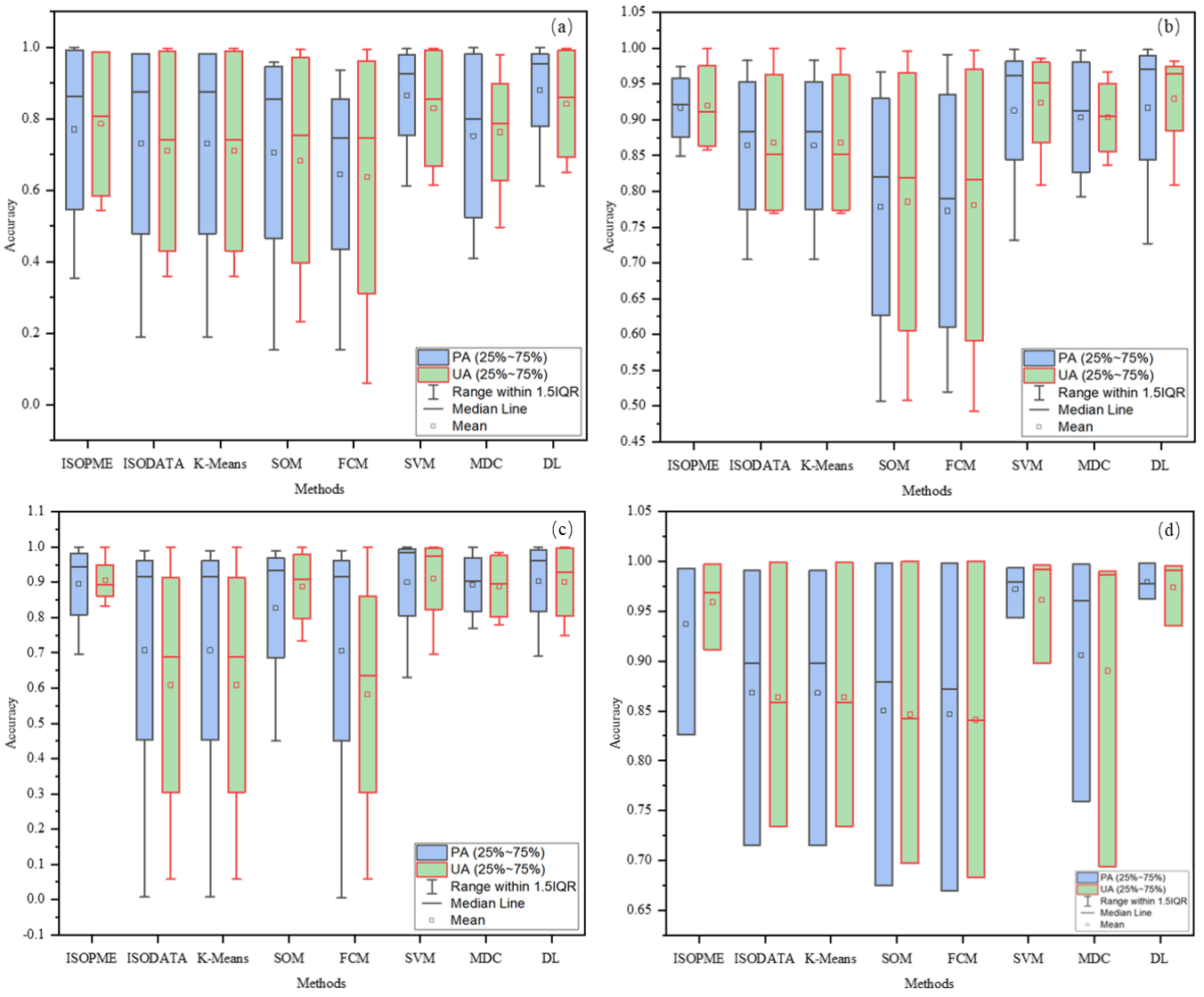


Fig. 14. PA and UA values for (a) first test area, (b) second test area, (c) third test area, and (d) fourth test area.

TABLE V
OA AND KC OF THE FOUR TEST AREAS

Test Areas		1st Test Area		2nd Test Area		3rd Test Area		4th Test Area	
Classification accuracy / Classification methods		OA	KC	OA	KC	OA	KC	OA	KC
Unsupervised classification	ISOPME	91.91%	0.84	91.48%	0.89	88.62%	0.84	96.08%	0.93
	ISODATA	89.43%	0.79	87.60%	0.83	76.36%	0.66	89.40%	0.83
	K-Means	89.43%	0.79	87.60%	0.83	76.36%	0.66	89.40%	0.83
	SOM	86.81%	0.73	78.33%	0.71	83.41%	0.76	87.97%	0.81
	FCM	70.19%	0.50	78.10%	0.71	71.10%	0.59	87.52%	0.80
Supervised classification	SVM	92.88%	0.87	91.35%	0.88	90.86%	0.88	97.39%	0.96
	MDC	88.43%	0.78	90.09%	0.87	90.37%	0.87	90.95%	0.86
Deep Learning (DL)		93.41%	0.88	91.70%	0.89	91.16%	0.88	98.21%	0.97

24.64%, and 9.78%. Therefore, the proposed ISOPME model demonstrates superior classification performance and higher classification accuracy. Overall, the ISOPME model can satisfy the classification requirements of multispectral RSIs.

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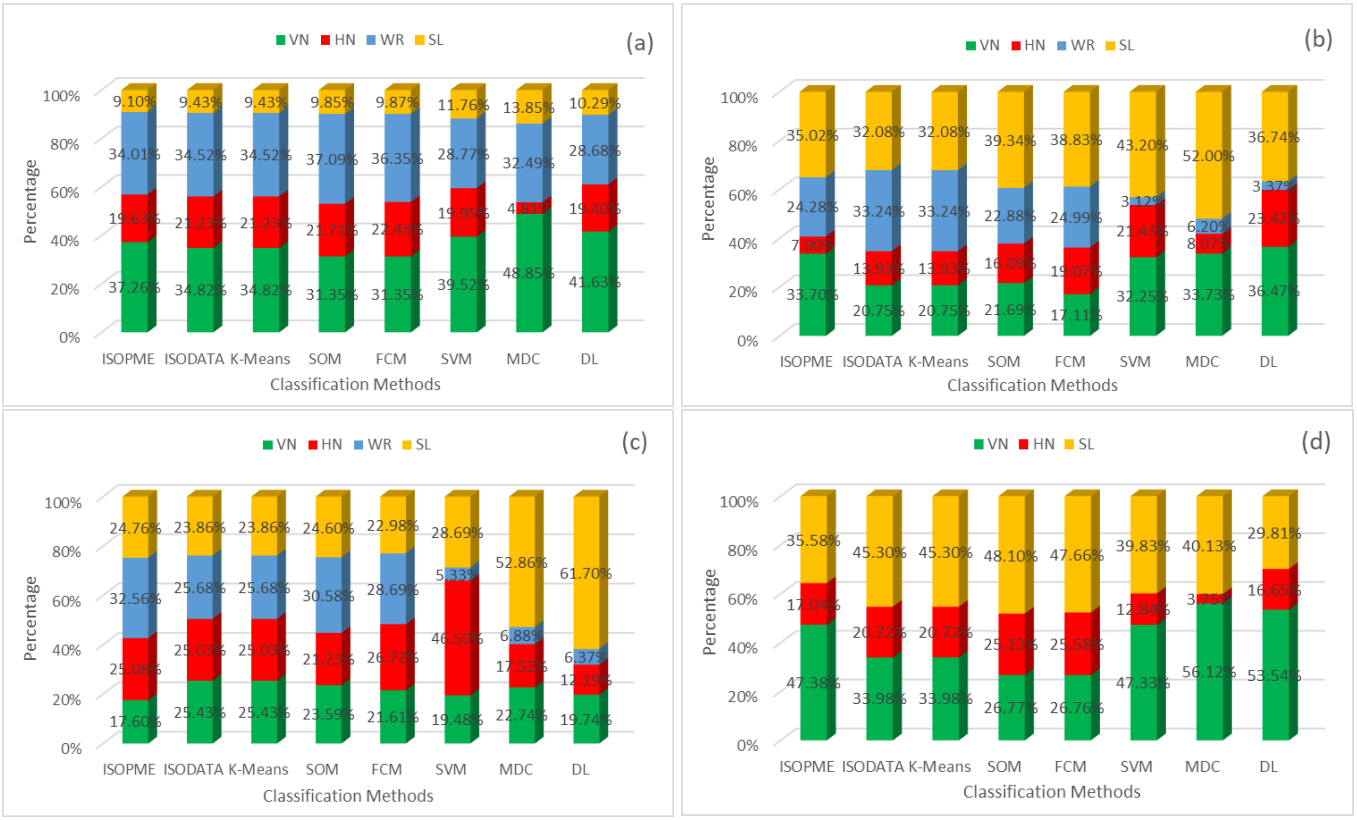


Fig. 15. Comparison of the proportion of classes about different classification methods from (a) first test area, (b) second test area, (c) third test area, and (d) fourth test area.

Furthermore, the confusion matrices in Tables I–IV of the four test areas indicate that: 1) the OA of the ISOPME model proposed in this article is 91.91%, 91.48%, 88.62%, and 96.08%, respectively, with the KC of 0.84, 0.89, 0.84, and 0.93; 2) the OA of the ISODATA model is 89.43%, 87.60%, 76.36%, and 89.40%, respectively, with the KC of 0.79, 0.83, 0.66, and 0.83; 3) the OA of the K-Means model is 89.43%, 87.60%,76.36%, and 89.40%, respectively, with the KC of 0.79, 0.83, 0.66, and 0.83; 4) the OA of the SOM model is 86.81%, 78.33%, 83.41%, and 87.97%, respectively, with the KC of 0.73, 0.71, 0.76, and 0.81; 5) the OA of the FCM model is 70.19%, 78.10%, 71.10%, and 87.52%, with the KC of 0.50, 0.71, 0.59, and 0.80, respectively; 6) the OA of the SVM model is 92.88%, 91.35%, 90.86%, and 97.39%, respectively, with the KC of 0.87, 0.88, 0.88, and 0.96; 7) the OA of the MDC model is 88.43%, 90.09%, 90.37%, and 90.95%, respectively, with the KC of 0.78, 0.87,0.87, and 0.86; and 8) the OA of the DL model is 93.41%, 91.70%, 91.16%, and 98.21%, respectively, with the KC of 0.88, 0.89, 0.88, and 0.97.

B. Sensitivity Analysis of ISOPME

To analyze the impact of different parameter settings on the classification accuracy of ISOPME, the first test area is selected for sensitivity analysis in this article. The iterations and MED thresholds are significant parameters that can be defined when executing the ISOPME model.

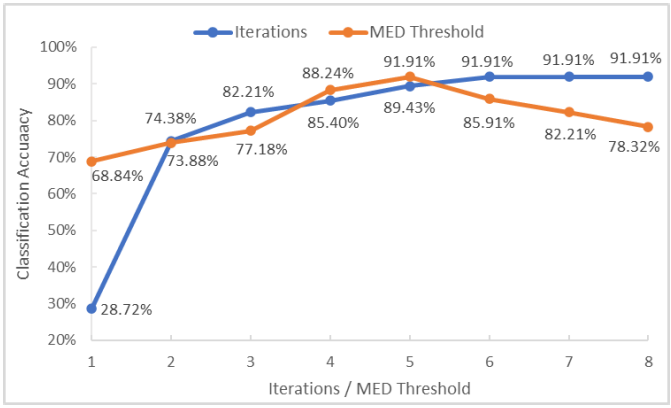


Fig. 16. Sensitivity analysis of iterations and MED threshold.

1) Sensitivity in Relation to the Iterations: The proposed ISOPME model conducts iteration operations based on the MED d and the PME coefficient μ to automatically split and merge cluster objects. Thus, it is essential to determine iterations prior to classification. The parameter settings of the iterations in the following experiment range from 1 to 8, and the sensitivity in relation to iterations is shown in Fig. 16.

From the analysis of Fig. 16, it can be concluded that the more the iterations, the higher the classification accuracy within limits, and the classification accuracy reaches its maximum value of 91.96% when the iterations reach 6. Therefore,

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the iterations must be larger than 5 while implementing the classification of the first test area in this article.

2) Sensitivity in Relation to the MED Threshold: Another key parameter is the MED threshold when splitting and merging cluster objects, which serves as a critical condition for classifying multiple objects into one with the MED. As shown in Fig. 16, the parameter settings of the MED threshold in the sensitivity analysis experiments range from 1 to 8.

It can be seen that the classification accuracy reaches its maximum value of 91.96% when the MED threshold is 5. In addition, it is impossible to achieve high classification accuracy for ISOPME if the MED threshold is too small or too large.

V. Conclusion

This article first proposes the PME theory to employ the quantum matrix entanglement characteristics concealed at the pixel level of RSIs, namely, PME, which effectively establishes a connection between the quantum states of pixels and their gray values. Then, the quantum PME coefficient μ is proposed to extract the entanglement relationship between pixel matrices with the 3-D orthogonal basis in the Hilbert space. Finally, an ISOPME model is developed for multispectral RSI classification.

The proposed ISOPME model based on the PME theory simultaneously exploits the spectral information of pixel matrices, the entanglement characteristics of pixel matrices, and the Euclidean distances between pixel matrices, achieving ideal classification performance and accurate classification results. In contrast, the traditional unsupervised classification models, such as ISODATA, K-Means, SOM, and FCM, only take into account the Euclidean distance features between pixels and do not reflect the rich spectral information between pixels, leading to low classification accuracy.

In the classification experiments of the four test areas, the ISOPME model proposed in this article is verified to have ideal accuracy and effectiveness for multispectral RSI classification. The following conclusions can be drawn: 1) compared with the traditional unsupervised classification models, ISODATA, K-Means, SOM, and FCM, the proposed ISOPME model achieves the classification accuracy of 91.91%, 91.48%, 88.62%, and 96.08% in the four test areas, with an average improvement of 8.96% and higher classification accuracy and 2) the classification accuracy of ISOPME proposed in this article has improved by an average of 7.68% compared with ISODATA, 7.68% compared with K-Means, 9.53% compared with SOM, 1.10% compared with SVM, and 1.60% compared with DL. However, to achieve the same desired classification accuracy, the SVM model requires more prior knowledge and the DL model requires more training samples than the ISOPME model.

References

[1] G. Zhou, F. Yang, and J. Xiao, "Study on pixel entanglement theory for imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409518, doi: 10.1109/TGRS.2022.3167569.

[2] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019, doi: 10.1109/MGRS.2019.2912563.

[3] Y. Sun, K. Deng, K. Ren, J. Liu, C. Deng, and Y. Jin, "Deep learning in statistical downscaling for deriving high spatial resolution gridded meteorological data: A systematic review," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 14–38, Feb. 2024, doi: 10.1016/j.isprsjprs.2023.12.011.

[4] P. Ghamisi et al., "New frontiers in spectral–spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 10–43, Sep. 2018, doi: 10.1109/MGRS.2018.2854840.

[5] W. B. Silva, C. C. Freitas, S. J. S. Sant'Anna, and A. C. Frery, "Classification of segments in PolSAR imagery by minimum stochastic distances between Wishart distributions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 3, pp. 1263–1273, Jun. 2013, doi: 10.1109/JSTARS.2013.2248132.

[6] M. Mafanya, P. Tsele, T. Zengeya, and A. Ramoelo, "An assessment of image classifiers for generating machine-learning training samples for mapping the invasive *Campuloclinium macrocephalum* (Less.) DC (pompom weed) using DESIS hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 188–200, Mar. 2022, doi: 10.1016/j.isprsjprs.2022.01.015.

[7] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017, doi: 10.1109/MGRS.2016.2616418.

[8] G. Zhang, R. Zhang, G. Zhou, and X. Jia, "Hierarchical spatial features learning with deep CNNs for very high-resolution remote sensing image classification," *Int. J. Remote Sens.*, vol. 39, no. 18, pp. 5978–5996, Sep. 2018, doi: 10.1080/01431161.2018.1506593.

[9] G. Zhou, R. Zhang, and D. Zhang, "Manifold learning co-location decision tree for remotely sensed imagery classification," *Remote Sens.*, vol. 8, no. 10, p. 855, Oct. 2016, doi: 10.3390/rs8100855.

[10] C. Li, Z. Shao, L. Zhang, X. Huang, and M. Zhang, "A comparative analysis of index-based methods for impervious surface mapping using multiseasonal Sentinel-2 satellite data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3682–3694, 2021, doi: 10.1109/JSTARS.2021.3067325.

[11] T. Gadhiya and A. K. Roy, "Superpixel-driven optimized Wishart network for fast PolSAR image classification using global k-means algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 97–109, Jan. 2020, doi: 10.1109/TGRS.2019.2933483.

[12] M. J. Cracknell, A. M. Reading, and P. de Caritat, "Multiple influences on regolith characteristics from continental-scale geophysical and mineralogical remote sensing data using self-organizing maps," *Remote Sens. Environ.*, vol. 165, pp. 86–99, Aug. 2015, doi: 10.1016/j.rse.2015.04.029.

[13] C.-C. Hung, S. Kulkarni, and B.-C. Kuo, "A new weighted fuzzy c-means clustering algorithm for remotely sensed image classification," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 543–553, Jun. 2011, doi: 10.1109/JSTSP.2010.2096797.

[14] J. Mäyrä et al., "Tree species classification from airborne hyperspectral and LiDAR data using 3D convolutional neural networks," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112322, doi: 10.1016/j.rse.2021.112322.

[15] P. Zhong, Z. Gong, S. Li, and C.-B. Schönlieb, "Learning to diversify deep belief networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3516–3530, Jun. 2017, doi: 10.1109/TGRS.2017.2675902.

[16] G. Liu, Y. Li, L. Jiao, Y. Chen, and R. Shang, "Multiobjective evolutionary algorithm assisted stacked autoencoder for PolSAR image classification," *Swarm Evol. Comput.*, vol. 60, Feb. 2021, Art. no. 100794, doi: 10.1016/j.swevo.2020.100794.

[17] S. Zhang, J. Wang, X. Tao, Y. Gong, and N. Zheng, "Constructing deep sparse coding network for image classification," *Pattern Recognit.*, vol. 64, pp. 130–140, Apr. 2017, doi: 10.1016/j.patcog.2016.10.032.

[18] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021, doi: 10.1109/MGRS.2021.3064051.

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[19] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020, doi: 10.1109/MGRS.2020.2979764.

[20] C. Ma, Q. Dai, and S. Liu, "A new method of remote sensing image segmentation based on PSO and Isodata," *Geomatics Inf. Sci. Wuhan Univ.*, vol. 37, no. 1, pp. 35–38, Jan. 2012, doi: 10.13203/j.whugis2012.01.006.

[21] A. Rydberg and G. Borgefors, "Integrated method for boundary delineation of agricultural fields in multispectral satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 11, pp. 2514–2520, Nov. 2001, doi: 10.1109/36.964989.

[22] Y. Zhong, L. Zhang, B. Huang, and P. Li, "An unsupervised artificial immune classifier for multi/hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 2, pp. 420–431, Feb. 2006, doi: 10.1109/TGRS.2005.861548.

[23] Y. Tarabalka, J. A. Benediktsson, and J. Chanussot, "Spectral–spatial classification of hyperspectral imagery based on partitional clustering techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 8, pp. 2973–2987, Aug. 2009, doi: 10.1109/TGRS.2009.2016214.

[24] D. Long and V. P. Singh, "An entropy-based multispectral image classification algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 12, pp. 5225–5238, Dec. 2013, doi: 10.1109/TGRS.2013.2272560.

[25] H. Jiao, Y. Zhong, and L. Zhang, "An unsupervised spectral matching classifier based on artificial DNA computing for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4524–4538, Aug. 2014, doi: 10.1109/TGRS.2013.2282356.

[26] R. Ulug and M. O. Karslioglu, "A new data-adaptive network design methodology based on the k-means clustering and modified ISODATA algorithm for regional gravity field modeling via spherical radial basis functions," *J. Geodesy*, vol. 96, no. 11, p. 91, Nov. 2022, doi: 10.1007/s00190-022-01681-2.

[27] S. Zhang, X. Li, Y. Ba, X. Lyu, M. Zhang, and M. Li, "Banana fusarium wilt disease detection by supervised and unsupervised methods from UAV-based multispectral imagery," *Remote Sens.*, vol. 14, no. 5, p. 1231, Mar. 2022, doi: 10.3390/rs14051231.

[28] Y. Shi, "The road of quantum entanglement: From Einstein to 2022 Nobel Prize in Physics," *Chin. J. Nature*, vol. 44, no. 6, pp. 455–465, 2022, doi: 10.3969/j.issn.0253-9608.2022.06.005.

[29] P. Zhang, "Quantum entanglement in the Sachdev—Ye—Kitaev model and its generalizations," *Frontiers Phys.*, vol. 17, no. 4, p. 43201, Aug. 2022, doi: 10.1007/s11467-022-1162-5.

[30] X. Wang and J. Pan, "The Nobel Prize in Physics 2022: Quantum entanglement," *Bull. Nat. Natural Sci. Found. China*, vol. 36, no. 6, pp. 928–930, 2022, doi: 10.16262/j.cnki.1000-8217.2022.06.003.

[31] P. Li, Y. Ling, and Z. Yu, "Generation rate of quantum gravity induced entanglement with multiple massive particles," *Phys. Rev. D, Part. Fields*, vol. 107, no. 6, Mar. 2023, Art. no. 064054, doi: 10.1103/physrevd.107.064054.

[32] C. Datta, T. Varun Kondra, M. Miller, and A. Streltsov, "Catalysis of entanglement and other quantum resources," *Rep. Prog. Phys.*, vol. 86, no. 11, Nov. 2023, Art. no. 116002, doi: 10.1088/1361-6633/acfbec.

[33] Y. Ruan, H. Chen, J. Tan, and X. Li, "Quantum computation for large-scale image classification," *Quantum Inf. Process.*, vol. 15, no. 10, pp. 4049–4069, Oct. 2016, doi: 10.1007/s11128-016-1391-z.

[34] H. Chen, F. Miao, and X. Shen, "Hyperspectral remote sensing image classification with CNN based on quantum genetic-optimized sparse representation," *IEEE Access*, vol. 8, pp. 99900–99909, 2020, doi: 10.1109/ACCESS.2020.2997912.

[35] F. Vernuccio et al., "Artificial intelligence in classical and quantum photonics," *Laser Photon. Rev.*, vol. 16, no. 5, May 2022, Art. no. 2100399, doi: 10.1002/lpor.202100399.

[36] A. Sebastianelli, D. A. Zaidenberg, D. Spiller, B. Le Saux, and S. L. Ullo, "On circuit-based hybrid quantum neural networks for remote sensing imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 565–580, 2022, doi: 10.1109/JSTARS.2021.3134785.

[37] P. Atchade-Adelomou and G. Alonso-Linaje, "Quantum-enhanced filter: QFilter," *Soft Comput.*, vol. 26, no. 15, pp. 7167–7174, Aug. 2022, doi: 10.1007/s00500-022-07190-w.

[38] P. Easom-McCaldin, A. Bouridane, A. Belatreche, R. Jiang, and S. Al-Maadeed, "Efficient quantum image classification using single qubit encoding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1472–1486, Feb. 2022, doi: 10.1109/TNNLS.2022.3179354.

[39] A. Zeguendry, Z. Jarir, and M. Quafafou, "Quantum machine learning: A review and case studies," *Entropy*, vol. 25, no. 2, p. 287, Feb. 2023, doi: 10.3390/e25020287.

[40] M. Nielsen and I. Chuang, "Introduction to quantum mechanics," in *Quantum Computation and Quantum Information*, 1st ed., Cambridge, U.K.: Cambridge Univ. Press, 2010, pp. 60–111.

[41] G. Aubrun and A. Müller-Hermes, "Annihilating entanglement between cones," *Commun. Math. Phys.*, vol. 400, no. 2, pp. 931–976, Jun. 2023, doi: 10.1007/s00220-022-04621-5.

[42] T. Mori, H. Manabe, and H. Matsueda, "Entanglement distillation toward minimal bond cut surface in tensor networks," *Phys. Rev. D, Part. Fields*, vol. 106, no. 8, Oct. 2022, Art. no. 086008, doi: 10.1103/physrevd.106.086008.

[43] D. M. Tong, L. C. Kwek, C. H. Oh, J.-L. Chen, and L. Ma, "Operator-sum representation of time-dependent density operators and its applications," *Phys. Rev. A, Gen. Phys.*, vol. 69, no. 5, May 2004, Art. no. 054102, doi: 10.1103/physreva.69.054102.

[44] P. Stelmachovič and V. Bužek, "Dynamics of open quantum systems initially entangled with environment: Beyond the Kraus representation," *Phys. Rev. A, Gen. Phys.*, vol. 64, no. 6, Nov. 2001, Art. no. 062106, doi: 10.1103/physreva.64.062106.

[45] A. Neven et al., "Symmetry-resolved entanglement detection using partial transpose moments," *npj Quantum Inf.*, vol. 7, no. 1, p. 152, Oct. 2021, doi: 10.1038/s41534-021-00487-y.

[46] J. R. Garrison and T. Grover, "Does a single eigenstate encode the full Hamiltonian?" *Phys. Rev. X*, vol. 8, no. 2, Apr. 2018, Art. no. 021026, doi: 10.1103/physrevx.8.021026.

[47] D. Shuai, Y. Liu, Q. Shuai, L. Huang, and Y. Dong, "Self-organizing data clustering based on quantum entanglement model," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Hangzhou, China, Jun. 2006, pp. 716–723.

[48] J. Gao, Q. Wu, C. Persson, and Z. Wang, "Irvsp: To obtain irreducible representations of electronic states in the VASP," *Comput. Phys. Commun.*, vol. 261, Apr. 2021, Art. no. 107760, doi: 10.1016/j.cpc.2020.107760.

[49] G. H. Low and I. L. Chuang, "Hamiltonian simulation by qubitization," *Quantum*, vol. 3, p. 163, Jul. 2019, doi: 10.22331/q-2019-07-12-163.

[50] N. Azamov, E. McDonald, F. Sukochev, and D. Zanin, "A dixmier trace formula for the density of states," *Commun. Math. Phys.*, vol. 377, no. 3, pp. 2597–2628, Aug. 2020, doi: 10.1007/s00220-020-03756-7.

[51] A. Rath et al., "Entanglement barrier and its symmetry resolution: Theory and experimental observation," *PRX Quantum*, vol. 4, no. 1, Feb. 2023, Art. no. 010318, doi: 10.1103/prxquantum.4.010318.

[52] Q. Zhang et al., "Applicability of quantum entanglement technology in geology," *Earth Sci. Frontiers*, vol. 26, no. 4, pp. 159–169, 2019.

[53] P. Nguyen, T. Devakul, M. G. Halbasch, M. P. Zaletel, and B. Swingle, "Entanglement of purification: From spin chains to holography," *J. High Energy Phys.*, vol. 2018, no. 1, pp. 1–41, Jan. 2018, doi: 10.1007/jhep01(2018)098.

[54] M. J. Gullans and D. A. Huse, "Dynamical purification phase transition induced by quantum measurements," *Phys. Rev. X*, vol. 10, no. 4, Oct. 2020, Art. no. 041020, doi: 10.1103/physrevx.10.041020.

[55] D. Shuai, P. Zhang, and L. Huang, "Self-organizing data clustering: A novel quantum particle approach," in *Proc. IEEE Int. Symp. Ind. Electron.*, Montreal, QC, Canada, Jun. 2006, pp. 2960–2965.

[56] F. Stephan, "The theory of self-adjoint operators in Hilbert spaces," in *Mathematical Foundations of Quantum Mechanics*, Princeton, NJ, USA: Princeton Univ. Press, 2015, pp. 15–32.

[57] M. K. Dhodhi, J. A. Saghri, I. Ahmad, and R. Ul-Mustafa, "D-ISODATA: A distributed algorithm for unsupervised classification of remotely sensed data on network of workstations," *J. Parallel Distrib. Comput.*, vol. 59, no. 2, pp. 280–301, Nov. 1999, doi: 10.1006/jpdc.1999.1573.

[58] S. J. Walsh et al., "QuickBird and hyperion data analysis of an invasive plant species in the galapagos islands of ecuador: Implications for control and land use management," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 1927–1941, May 2008, doi: 10.1016/j.rse.2007.06.028.

[59] C. Yang, Z. Wang, S.-K. Oh, W. Pedrycz, and B. Yang, "Ensemble fuzzy radial basis function neural networks architecture driven with the aid of multi-optimization through clustering techniques and polynomial-based learning," *Fuzzy Sets Syst.*, vol. 438, pp. 62–83, Jun. 2022, doi: 10.1016/j.fss.2021.06.014.

[60] O. Albatayneh, L. Forslöf, and K. Ksaibati, "Image retraining using TensorFlow implementation of the pretrained inception-v3 model for evaluating gravel road dust," *J. Infrastructure Syst.*, vol. 26, no. 2, Jun. 2020, Art. no. 04020014, doi: 10.1061/(asce)is.1943-555x.0000545.

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